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Sound collecting method, electronic device, and system

Abstract

A sound collecting system includes a first electronic device, a left-ear TWS earphone, and a right-ear TWS earphone. Microphones of the first electronic device, the left-ear TWS earphone, and the right-ear TWS earphone form a first microphone array. When the first electronic device detects a user operation of starting video recording, the first microphone array performs sound collecting to obtain a joint microphone array signal. The first electronic device performs noise reduction processing on the joint microphone array signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a U.S. National Stage of International Patent Application No. PCT/CN2021/137406 filed on Dec. 13, 2021, which claims priority to Chinese Patent Application No. 202011593358.1 filed on Dec. 29, 2020. Both of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) This application relates to the field of terminal technologies, and in particular, to a sound collecting method, an electronic device, and a system.

BACKGROUND

(3) With the development of Bluetooth technologies, a true wireless stereo (true wireless stereo, TWS) headset can transmit audio signals with electronic devices such as a mobile phone and a tablet computer more efficiently and with high quality, and is favored by more users. In video recording scenarios such as live broadcast and video blog (video flog, vlog) shooting, a user may collect sound by using a microphone of a TWS headset.

(4) However, when a video is recorded in a noisy environment, an existing sound pickup noise reduction solution used by a TWS headset is affected by a limitation of a quantity of microphones in the headset and a user wearing posture. The TWS headset has a poor effect in suppressing ambient noise and enhancing the target sound.

SUMMARY

(5) This application provides a sound collecting method, an electronic device, and a system, so that when a user wears a wireless headset and uses a first electronic device to record a video, the wireless headset and a microphone of the first electronic device can be used to perform sound collecting. The first electronic device may perform noise reduction processing on audio signals obtained by using a plurality of microphones, to improve sound quality in a recorded video.

(6) According to a first aspect, this application provides a sound collecting method. The method may be applied to a first electronic device with a first microphone, a left-ear wireless earphone with a second microphone, and a right-car wireless earphone with a third microphone. The first electronic device is connected to the left-ear wireless earphone and the right-ear wireless earphone through wireless communication. The method may include: The first electronic device collects a face image. The first electronic device may determine relative locations of the first microphone, the second microphone, and the third microphone based on the face image and posture information of the first electronic device. The first electronic device may obtain a first audio signal of the first microphone, a second audio signal of the second microphone, and a third audio signal of the third microphone. The first electronic device may perform noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations.

(7) The first microphone, the second microphone, and the third microphone may form a first microphone array. The first microphone may include one or more microphones in the first electronic device. The second microphone may include one or more microphones in the left-ear wireless earphone. The third microphone may include one or more microphones in the right-ear wireless earphone.

(8) It can be learned from a location of each microphone in the first microphone array that a near field area formed by the first microphone array includes an area in which a user wearing a TWS headset and the first electronic device **100** are located. Compared with a microphone array in a single TWS earphone, the first microphone array has a larger size and a stronger spatial resolution capability, and can more accurately distinguish a target sound in the near field area and the ambient noise from a far field area. In this way, the first electronic device **100** can better enhance the target sound and suppress the ambient noise when performing spatial filtering on the audio signal collected by the first microphone array, to improve sound quality in a recorded video.

(9) In addition, adding the microphone in the first electronic device **100** to the microphone array can reduce impact of a posture of wearing the TWS headset by the user on enhancing the target sound and reducing the ambient noise by the first electronic device **100**.

(10) The target sound may be a sound whose sound source is located in the near field area of the first microphone array. The target sound may include a voice of the user and a sound of playing a musical instrument by the user.

(11) With reference to the first aspect, in some embodiments, before performing noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations, the first electronic device may further perform delay alignment on the first audio signal, the second audio signal, and the third audio signal.

(12) Specifically, the first electronic device may emit an alignment sound. The aligned sound is

obtained by performing digital-to-analog conversion on an alignment signal. The first electronic device may perform delay correlation detection on a first alignment signal part of the first audio signal, a second alignment signal part of the second audio signal, and a third alignment signal part of the third audio signal, to determine delay lengths between the first audio signal, the second audio signal, and the third audio signal. The first electronic device may perform delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the delay lengths.
(13) The alignment signal may be an audio signal with a frequency higher than 20000 Hz. An audible frequency band of a human ear ranges from 20 Hz to 20000 Hz. If the frequency of the alignment signal is higher than 20000 Hz, and the alignment sound is not heard by the user. This can avoid interference of the alignment sound to the user.
(14) The delay alignment may determine data that is collected at a same time point in the first audio signal, the second audio signal, and the third audio signal, to reduce impact of a delay difference on noise reduction processing performed by the first electronic device.
(15) In a possible implementation, the first electronic device may use any one of the first audio signal, the second audio signal, and the third audio signal as a reference audio signal, and perform delay alignment on another audio signal and the reference audio signal. For example, the first electronic device may use the first audio signal as a reference audio signal. The first electronic device may perform high-pass filtering on the first audio signal, the second audio signal, and the third audio signal, to obtain alignment signal parts in these audio signals. The first electronic device may perform delay processing of different time lengths on the alignment signal parts in the second audio signal and the third audio signal, to determine delay lengths at which a correlation between the alignment signal parts of the second audio signal and the third audio signal and the alignment signal part of the first audio signal is the highest. In this way, the first electronic device may determine the delay lengths of the second audio signal and the third audio signal relative to the first audio signal.
(16) In a possible implementation, the first electronic device may generate the alignment signal in a preset time period (for example, a preset time period before video recording starts or a preset time period after video recording starts) for starting video recording, and emit the alignment sound. The first electronic device may determine the delay lengths between the first audio signal, the second audio signal, and the third audio signal based on the alignment signal of the preset time period. The delay lengths generally do not change or change Very little in a video recording process of the first electronic device. In a video recording process after the preset time period, the first electronic device may perform delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the delay lengths. In addition, the first electronic device may stop generating the alignment signal, to save power consumption of the first electronic device.
(17) With reference to the first aspect, in some embodiments, the relative locations of the first microphone, the second microphone, and the third microphone may include coordinates of the first microphone, the second microphone, and the third microphone in a world coordinate system.
(18) The first electronic device may determine a first conversion relationship between a standard head coordinate system and a first electronic device coordinate system based on a correspondence between coordinates of a first human face key point in the standard head coordinate system and coordinates of the first human face key point in a face image coordinate system. The standard head coordinate system is determined based on a standard head model. The first electronic device may store coordinates of each key point in the standard head model in the standard head coordinate system. The first electronic device may determine coordinates of a left ear and a right ear in the standard head model in the first electronic device coordinate system based on the first conversion relationship and coordinates of the left ear and the right ear in the standard head model in the standard head coordinate system. The coordinates of the left ear and the right ear in the standard head model in the first electronic device coordinate system are respectively coordinates of the second microphone and the third microphone in the first electronic device coordinate system. The

first electronic device may determine a second conversion relationship between the first electronic device coordinate system and the world coordinate system based on the posture information of the first electronic device. The first electronic device may be based on the second conversion relationship, and coordinates (that is, the foregoing relative locations) of the first microphone, the second microphone, and the third microphone in the first electronic device coordinate system.
(19) The standard head coordinate system may be a three-dimensional coordinate system established by using a location of a nose tip in the standard head model as an origin, using a direction perpendicular to a face as a direction of an x-axis, using a horizontal direction parallel to the face as a direction of a y-axis, and using a vertical direction parallel to the face as a direction of a z-axis.
(20) The first face key point may include any plurality of key points of an area in which the face is located in the face image, for example, a key point in an area in which a forehead area is located, a key point in an area in which a cheek is located, and a key point in an area in which a lip is located.
(21) The posture information may be determined by the first electronic device by using a posture sensor (for example, an acceleration sensor or a gyroscope sensor).
(22) A placement posture of the first electronic device is unknown and time-varying. The target sound that needs to be collected by the first microphone array includes the voice of the user, and may include the sound of playing a musical instrument by the user. Compared with coordinates of each microphone in the first microphone array in the electronic device coordinate system, that the first electronic device uses the coordinates of each microphone in the first microphone array in the world coordinate system in the spatial filtering process can better improve an effect of enhancing the target sound and reducing the ambient noise.
(23) With reference to the first aspect, in some embodiments, a method for performing, by the first electronic device, noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations may specifically include: The first electronic device may perform voice activity detection on the first audio signal, the second audio signal, and the third audio signal based on the relative locations. The voice activity detection is used to determine a frequency point of a target sound signal and a frequency point of an ambient noise signal in the first audio signal, the second audio signal, and the third audio signal. The first electronic device may update a noise spatial characteristic of ambient noise based on the frequency point of the target sound signal and the frequency point of the ambient noise signal. The noise spatial characteristic indicates spatial distribution of the ambient noise. The spatial distribution of the ambient noise includes a direction and energy of the ambient noise. The first electronic device may determine a target steering vector of the first audio signal, the second audio signal, and the third audio signal based on the relative locations. The target steering vector may indicate a direction of the target sound signal. The first electronic device may determine a spatial filter based on the noise spatial characteristic and the target steering vector, and perform spatial filtering on the first audio signal, the second audio signal, and the third audio signal by using the spatial filter, to enhance the target sound signal in the first audio signal, the second audio signal, and the third audio signal and suppress the ambient noise signal in the first audio signal, the second audio signal, and the third audio signal.
(24) With reference to the first aspect, in some embodiments, the left-ear wireless earphone may perform wearing detection. The wearing detection may be used to determine whether the left-ear wireless earphone is in an in-ear state. When the left-ear wireless earphone is in the in-ear state, the left-ear wireless earphone obtains the second audio signal by using the second microphone. The right-ear wireless earphone may perform the wearing detection. When the right-ear wireless earphone is in the in-ear state, the right-ear wireless earphone obtains the third audio signal by using the third microphone.
(25) In a possible implementation, when detecting a user operation used to start video recording, the first electronic device may query a wearing detection result from the left-ear wireless earphone

and the right-ear wireless earphone. When it is determined that both the left-ear wireless earphone and the right-ear wireless earphone are in the in-ear state, the first electronic device may turn on the first microphone, and send a microphone turn-on instruction to the left-ear wireless earphone and the right-ear wireless earphone. When receiving the microphone turn-on instruction from the first electronic device, the left-ear wireless earphone and the right-ear wireless earphone may respectively turn on the second microphone and the third microphone.
(26) In another possible implementation, the left-ear wireless earphone may turn on the second microphone when performing the wearing detection and determining that the left-ear wireless earphone is in the in-ear state. The right-ear wireless earphone may turn on the third microphone when performing wearing detection and determining that the right-ear wireless earphone is in the in-ear state. That is, the first electronic device may not need to send the microphone turn-on instruction to the left-ear wireless earphone and the right-ear wireless earphone.
(27) With reference to the first aspect, in some embodiments, the first electronic device may mix a first video and a fourth audio signal that are collected in a first time period. The fourth audio signal may be an audio signal obtained after the noise reduction processing is performed on the first audio signal, the second audio signal, and the third audio signal. The first audio signal, the second audio signal, and the third audio signal are respectively obtained in the first time period by using the first microphone, the second microphone, and the third microphone.
(28) In some embodiments, a microphone of one of the left-ear wireless earphone and the right-ear wireless earphone may form a microphone array with the microphone of the first electronic device. The microphone array may also be applicable to the sound collecting method provided in embodiments of this application. For example, in a scenario in which the user wears the left-ear wireless earphone and the right-ear wireless earphone and uses the first electronic device for live broadcast, the first microphone of the first electronic device and the second microphone of the left-ear wireless earphone may perform sound collecting. The first electronic device may obtain the first audio signal by using the first microphone. The left-ear wireless earphone may obtain the second audio signal by using the second microphone. The left-ear wireless earphone may send the second audio signal to the first electronic device. The first electronic device may perform noise reduction processing on the first audio signal and the second audio signal, to obtain an audio signal in a live video.
(29) A near field area of the microphone array formed by the microphone of one earphone and the microphone of the first electronic device may still include a location of a sound source of a target sound, such as a user voice or a sound of playing a musical instrument by the user. In addition, using the microphone of one earphone can reduce power consumption of the headset.
(30) According to a second aspect, this application further provides a sound collecting method. The method is applied to a first electronic device with a first microphone. The first electronic device is connected to a left-ear wireless earphone with a second microphone and a right-ear wireless earphone with a third microphone through wireless communication. The method may include: The first electronic device collects a face image. The first electronic device determines relative locations of the first microphone, the second microphone, and the third microphone based on the face image and posture information of the first electronic device. The first electronic device obtains a first audio signal of the first microphone, a second audio signal of the second microphone, and a third audio signal of the third microphone. The first electronic device performs noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations.
(31) The first microphone, the second microphone, and the third microphone may form a first microphone array. The first microphone may include one or more microphones in the first electronic device. The second microphone may include one or more microphones in the left-ear wireless earphone. The third microphone may include one or more microphones in the right-ear wireless earphone.

(32) It can be learned from a location of each microphone in the first microphone array that a near field area formed by the first microphone array includes an area in which a user wearing a TWS headset and the first electronic device **100** are located. Compared with a microphone array in a single TWS earphone, the first microphone array has a larger size and a stronger spatial resolution capability, and can more accurately distinguish a target sound in the near field area and ambient noise from a far field area. In this way, the first electronic device **100** can better enhance the target sound and suppress the ambient noise when performing spatial filtering on the audio signal collected by the first microphone array, to improve sound quality in a recorded video.
(33) In addition, adding the microphone in the first electronic device **100** to the microphone array can reduce impact of a posture of wearing the TWS headset by the user on enhancing the target sound and reducing the ambient noise by the first electronic device **100**.
(34) The target sound may be a sound whose sound source is located in the near field area of the first microphone array. The target sound may include a voice of the user and a sound of playing a musical instrument by the user.
(35) With reference to the first aspect, in some embodiments, before performing noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations, the first electronic device may further perform delay alignment on the first audio signal, the second audio signal, and the third audio signal.
(36) Specifically, the first electronic device may emit an alignment sound. The aligned sound is obtained by performing digital-to-analog conversion on an alignment signal. The first electronic device may perform delay correlation detection on a first alignment signal part of the first audio signal, a second alignment signal part of the second audio signal, and a third alignment signal part of the third audio signal, to determine delay lengths between the first audio signal, the second audio signal, and the third audio signal. The first electronic device may perform delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the delay lengths.
(37) The alignment signal may be an audio signal with a frequency higher than 20000 Hz. An audible frequency band of a human ear ranges from 20 Hz to 20000 Hz. If the frequency of the alignment signal is higher than 20000 Hz, and the alignment sound is not heard by the user. This can avoid interference of the alignment sound to the user.
(38) The delay alignment may determine data that is collected at a same time point in the first audio signal, the second audio signal, and the third audio signal, to reduce impact of a delay difference on noise reduction processing performed by the first electronic device.
(39) With reference to the second aspect, in some embodiments, the relative locations of the first microphone, the second microphone, and the third microphone may include coordinates of the first microphone, the second microphone, and the third microphone in a world coordinate system.
(40) The first electronic device may determine a first conversion relationship between a standard head coordinate system and a first electronic device coordinate system based on a correspondence between coordinates of a first human face key point in the standard head coordinate system and coordinates of the first human face key point in a face image coordinate system. The standard head coordinate system is determined based on a standard head model. The first electronic device may store coordinates of each key point in the standard head model in the standard head coordinate system. The first electronic device may determine coordinates of a left ear and a right ear in the standard head model in the first electronic device coordinate system based on the first conversion relationship and coordinates of the left ear and the right ear in the standard head model in the standard head coordinate system. The coordinates of the left ear and the right ear in the standard head model in the first electronic device coordinate system are respectively coordinates of the second microphone and the third microphone in the first electronic device coordinate system. The first electronic device may determine a second conversion relationship between the first electronic device coordinate system and the world coordinate system based on the posture information of the first electronic device. The first electronic device may be based on the second conversion

relationship, and coordinates (that is, the foregoing relative locations) of the first microphone, the second microphone, and the third microphone in the first electronic device coordinate system.
(41) A placement posture of the first electronic device is unknown and time-varying. The target sound that needs to be collected by the first microphone array includes the voice of the user, and may include the sound of playing a musical instrument by the user. Compared with coordinates of each microphone in the first microphone array in the electronic device coordinate system, that the first electronic device uses the coordinates of each microphone in the first microphone array in the world coordinate system in the spatial filtering process can better improve an effect of enhancing the target sound and reducing the ambient noise.
(42) With reference to the second aspect, in some embodiments, a method for performing, by the first electronic device, noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations may specifically include: The first electronic device may perform voice activity detection on the first audio signal, the second audio signal, and the third audio signal based on the relative locations. The voice activity detection is used to determine a frequency point of a target sound signal and a frequency point of an ambient noise signal in the first audio signal, the second audio signal, and the third audio signal. The first electronic device may update a noise spatial characteristic of ambient noise based on the frequency point of the target sound signal and the frequency point of the ambient noise signal. The noise spatial characteristic indicates spatial distribution of the ambient noise. The spatial distribution of the ambient noise includes a direction and energy of the ambient noise. The first electronic device may determine a target steering vector of the first audio signal, the second audio signal, and the third audio signal based on the relative locations. The target steering vector may indicate a direction of the target sound signal. The first electronic device may determine a spatial filter based on the noise spatial characteristic and the target steering vector, and perform spatial filtering on the first audio signal, the second audio signal, and the third audio signal by using the spatial filter, to enhance the target sound signal in the first audio signal, the second audio signal, and the third audio signal and suppress the ambient noise signal in the first audio signal, the second audio signal, and the third audio signal.
(43) With reference to the second aspect, in some embodiments, the first electronic device may mix a first video and a fourth audio signal that are collected in a first time period. The fourth audio signal may be an audio signal obtained after the noise reduction processing is performed on the first audio signal, the second audio signal, and the third audio signal. The first audio signal, the second audio signal, and the third audio signal may all be audio signals collected in the first time period.
(44) According to a third aspect, this application provides an electronic device. The electronic device may include a communication apparatus, a camera, a microphone, a memory, and a processor. The communication apparatus may be configured to establish a communication connection to a wireless headset. The camera may be configured to collect an image. The microphone may be configured to perform sound collecting. The memory may be configured to store a standard head coordinate system and a computer program. The processor may be configured to invoke the computer program, so that the electronic device performs any possible implementation of the second aspect.
(45) According to a fourth aspect, this application provides a computer storage medium, including instructions. When the instructions are run on an electronic device, the electronic device performs any possible implementation of the first aspect or the electronic device performs any possible implementation of the second aspect.
(46) According to a fifth aspect, an embodiment of this application provides a chip. The chip is applied to an electronic device, the chip includes one or more processors, and the processor is configured to invoke computer instructions, so that the electronic device performs any one possible implementation of the first aspect, or the electronic device performs any possible implementation of the second aspect.

(47) According to a sixth aspect, an embodiment of this application provides a computer program product including instructions. When the computer program product runs on a device, the foregoing electronic device performs any possible implementation of the first aspect, or the electronic device performs any possible implementation of the second aspect.
(48) It may be understood that the electronic device provided in the third aspect, the computer storage medium provided in the fourth aspect, the chip provided in the fifth aspect, and the computer program product provided in the sixth aspect are all configured to perform the method provided in embodiments of this application. Therefore, for beneficial effects that can be achieved, refer to beneficial effects in a corresponding method. Details are not described herein again.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1**A is a schematic diagram of a sound collecting scenario according to an embodiment of this application;
(2) FIG. **1**B is a schematic diagram of a spatial resolution capability of a microphone array according to an embodiment of this application;
(3) FIG. **2**A is a schematic diagram of another sound collecting scenario according to an embodiment of this application;
(4) FIG. **2**B is a schematic diagram of a spatial resolution capability of another microphone array according to an embodiment of this application;
(5) FIG. **3**A to FIG. **3**G are schematic diagrams of some sound collecting scenarios according to an embodiment of this application;
(6) FIG. **4** is a schematic diagram of a structure of a sound collecting system according to an embodiment of this application;
(7) FIG. **5**A and FIG. **5**B are schematic diagrams of a method for performing delay alignment on an audio signal according to an embodiment of this application;
(8) FIG. **6**A to FIG. **6**C are some schematic diagrams of coordinate conversion according to an embodiment of this application;
(9) FIG. **7**A is a flowchart of a method for turning on a first microphone array according to an embodiment of this application;
(10) FIG. **7**B is a flowchart of a sound collecting method according to an embodiment of this application; and
(11) FIG. **8** is a schematic diagram of a structure of a first electronic device **100** according to an embodiment of this application.
DESCRIPTION OF EMBODIMENTS
(12) Technical solutions according to embodiments of this application are clearly and completely described in the following with reference to accompanying drawings. In the descriptions of embodiments of this application, unless otherwise specified, "/" indicates "or". For example, A/B may indicate A or B. The term "and/or" in this specification merely describes an association relationship for describing associated objects, and indicates that three relationships may exist. For example, A and/or B may indicate the following three cases: Only A exists, both A and B exist, and only B exists. In addition, in the descriptions of embodiments of this application, "a plurality of" means two or more.
(13) The following terms "first" and "second" are merely intended for a purpose of description, and shall not be understood as an indication or implication of relative importance or implicit indication of a quantity of indicated technical features. Therefore, a feature limited by "first" or "second" may explicitly or implicitly include one or more features. In the descriptions of embodiments of this application, unless otherwise specified, "a plurality of" means two or more than two.

(14) In some embodiments, a first electronic device (such as a mobile phone or a tablet computer) establishes a communication connection to a TWS headset. In a scenario such as video recording or a voice call, the first electronic device may collect a sound by using a microphone of the TWS headset. As shown in FIG. **1**A, the TWS headset may collect a sound by using a microphone array of one of a left-ear TWS earphone **201** or a right-ear TWS earphone **202**. The sound collected by the TWS headset usually includes ambient noise. The foregoing ambient noise mainly comes from a far field area of a microphone array in a single TWS earphone. The ambient noise may be interference noise in a plurality of directions such as front interference noise, rear interference noise, left interference noise, and right interference noise of the TWS earphone. The first electronic device may perform spatial filtering on an audio signal collected by the microphone array in the single TWS headset, to suppress ambient noise and enhance a target sound in a near field area. However, a size of the TWS headset is limited. The near field area of the microphone array in the single TWS earphone is very small. A sound making location of a user mouth is already located in the far field area of the microphone array in the single TWS earphone. In this case, it is difficult for the first electronic device to distinguish between user voice and ambient noise in a same direction in the audio signal collected by the microphone array in the single TWS earphone.
(15) FIG. **1**B shows an example of a spatial resolution capability of a microphone array of a single TWS earphone. Because a distance between microphones in the microphone array of the single TWS earphone is short, a size of the microphone array is insufficient, and the spatial resolution capability of the microphone array of the single TWS earphone is weak. A near field target source of the microphone array of the single TWS earphone often cannot include a sound source from a user voice. When a first electronic device performs spatial filtering, it is difficult to effectively enhance the user voice by enhancing the near field target source. In addition, an orientation of the microphone array of the single TWS earphone is easily affected by a user wearing posture, which leads to a deviation of a beam main lobe direction from a user mouth direction during spatial filtering. A sound pickup capability of the microphone array of the single TWS earphone is further weakened. In addition, the microphone array of the single TWS earphone is limited by a size constraint, and a spatial spectral peak of a determined spatial spectrum of the sound source is wide. A target source signal is easily shielded by an ambient noise signal. It is difficult for the microphone array in the single TWS earphone to accurately distinguish the target source from ambient noise. When the first electronic device performs spatial filtering on an audio signal collected by the microphone array in the single TWS earphone, an effect of suppressing the ambient noise and enhancing the user voice by the first electronic device is poor. In a video recorded by the first electronic device, ambient noise is large, and a user voice is not clear enough.
(16) This application provides a sound collecting method. According to the method, when a communication connection is established between a TWS headset and a first electronic device **100** (such as a mobile phone or a tablet computer), and the first electronic device **100** records a video, a target sound may be enhanced, and ambient noise may be suppressed, to improve sound quality in the recorded video. Specifically, as shown in FIG. **2**A, a microphone of the first electronic device **100** and microphones of left-ear and right-ear TWS earphones may form a first microphone array. The target sound is a sound of a sound source in a near field area of the first microphone array. The target sound may be, for example, a user voice or a sound of playing a musical instrument by the user. A size of the first microphone array is large. The near field area of the first microphone array may include an area formed by locations at which a user and the first electronic device **100** are located. The foregoing ambient noise is mainly a sound of the sound source in a far field area of the first microphone array.
(17) FIG. **2**B shows an example of a spatial resolution capability of the first microphone array. The near field area formed by the first microphone array combined with the microphone of the first electronic device **100** is larger than a near field area of a microphone array of a single TWS earphone, and is less affected by a user wearing posture. The first microphone array has a stronger

spatial resolution capability. The first microphone array may more accurately distinguish a direction and a distance between each sound source and the first microphone array in a video recording process of the first electronic device. When a spatial energy spectrum of ambient noise from any direction of the far field area reaches the near field area, obvious attenuation occurs, and the first electronic device **100** can better distinguish the ambient noise sound from a target sound. The first electronic device may perform spatial filtering on an audio signal collected by the first microphone array, and suppress the ambient noise sound and enhance the target sound based on directions and distances of the target sound and the ambient noise. This reduces impact of the ambient noise on a video recorded by the first electronic device, and improves sound quality in the recorded video.

(18) The schematic diagram of the spatial resolution capability shown in FIG. **2**B is merely used to explain this application, and should not constitute a limitation on the spatial resolution capability of the first microphone array.

(19) In some embodiments, there may be one or more microphones of the first electronic device **100** to form the first microphone array. There may also be one or more microphones of the left-ear and right-ear TWS earphones to form the first microphone array. That is, the first microphone array includes at least three microphones. The three microphones are respectively from the first electronic device **100**, a left-ear TWS earphone **201**, and a right-ear TWS earphone **202**.

(20) In addition, a headset is not limited to a TWS headset, and may be another type of headset, for example, a head mounted headset. When the headset establishes a communication connection to the first electronic device **100**, and the first electronic device records a video, the microphone of the first electronic device **100** and the microphones of the headset may form the first microphone array to collect a sound.

(21) The first electronic device **100** may be an electronic device with a camera, such as a mobile phone, a tablet computer, a notebook computer, a television, an ultra-mobile personal computer (ultra-mobile personal computer, UMPC), a handheld computer, a netbook, or a personal digital assistant (personal digital assistant, PDA). A specific type of the first electronic device **100** is not limited in this embodiment of this application.

(22) In subsequent embodiments of this application, a microphone array including one microphone in the first electronic device **100**, one microphone in the left-ear TWS earphone **201**, and one microphone in the right-ear TWS earphone **202** is specifically used as an example to describe the sound collecting method provided in this application.

(23) The following describes a near field area and a far field area of a microphone array mentioned in embodiments of this application.

(24) Based on a distance between a sound source and the microphone array, a sound field model can be classified into a near field model and a far field model. A sound field model in the near field area of the microphone array is the near field model. In the near field model, a response of the microphone array is related to a direction of an incident sound source, and is related to a distance of the incident sound source. A sound wave is a spherical wave. A sound field model in the far field area of the microphone array is the far field model. In the far field model, a response of the microphone array is only related to a direction of an incident sound source, and is not related to a distance of the incident sound source. A sound wave is a plane wave.

(25) A size of the near field area of the microphone array is positively correlated with a size of the microphone array. There is no absolute standard for dividing the near field area and the far field area. In some embodiments, an area whose distance from a central reference point of the microphone array is greater than a wavelength of a sound signal is the far field area of the microphone array. Otherwise, an area whose distance from a central reference point of the microphone array is less than a wavelength of a sound signal is the far field area of the microphone array is the near field area of the microphone array. A manner of dividing the near field area and the far field area of the microphone array is not limited in this embodiment of this application.

(26) In this embodiment of this application, a size of a first microphone array is large, so that a location of a sound source of a sound that needs to be enhanced (for example, a user voice) can be better included in a near field area of the first microphone array. When performing spatial filtering on an audio signal collected by the first microphone array, a first electronic device may better distinguish a sound source of a target sound from a sound source of ambient noise by using differences of directions and distances. The sound source of the target sound may be a sound source located in the near field area of the first microphone array. The sound source of the ambient noise may be a sound source located in a far field area of the first microphone array.
(27) The sound collecting method of this application can be applied to a scenario in which the first electronic device **100** is connected to the TWS headset and a front camera is turned on for live broadcast, vlog shooting and other video recording. In this case, the target sound may include a user voice and a sound of playing a musical instrument by a user.
(28) The following describes a sound collecting scenario in an embodiment of this application.
(29) In some embodiments, the first electronic device **100** may detect a user operation of enabling a sound source enhancement function. In a video recording scenario such as live broadcast or vlog shooting, the first electronic device **100** may use the microphone of the first electronic device **100**, the microphone of the left-ear TWS earphone **201**, and the microphone of the right-ear TWS earphone **202** to form the first microphone array, to collect a sound. The first electronic device **100** may enhance a target sound from a user based on audio collected by the first microphone array, and suppress ambient noise in the target sound, to improve sound quality in a recorded video.
(30) For example, FIG. **3**A to FIG. **3**G are schematic diagrams of a sound collecting scenario according to an embodiment of this application.
(31) As shown in FIG. **3**A, each of a left-ear TWS earphone **201** and a right-ear TWS earphone **202** establishes a Bluetooth connection to a first electronic device **100**. In a case in which a TWS headset establishes a Bluetooth connection to the first electronic device **100**, the first electronic device **100** may send through Bluetooth, audio to be played to the TWS headset. An audio output apparatus of the TWS headset, for example, a speaker, may play the received audio. The TWS headset may send, to the first electronic device **100** through Bluetooth, audio collected by using an audio input apparatus (for example, a microphone).
(32) The first electronic device **100** may be provided with a camera **193**. The camera **193** may include a front camera and a rear camera. The front camera of the first electronic device **100** may be shown in FIG. **3**A. A quantity of cameras **193** of the first electronic device **100** is not limited in this embodiment of this application.
(33) The first electronic device **100** may display a user interface **310** shown in FIG. **3**A. The user interface **310** may include a camera application icon **311**. In response to a user operation performed on the camera application icon **311**, the first electronic device **100** may start a camera application. For example, the first electronic device **100** may turn on the front camera, and display a user interface **320** shown in FIG. **3**B.
(34) In response to a user operation performed on the camera application icon **311**, the first electronic device **100** may further turn on the rear camera, or turn on the front camera and the rear camera. This is not limited in this embodiment of this application.
(35) The user interface **310** may further include more content. This is not limited in this embodiment of this application.
(36) As shown in FIG. **3**B, the user interface **320** may include a flash control **321**, a setting control **322**, a preview box **323**, a camera mode option **324**, a gallery shortcut button **325**, a video recording start control **326**, and a camera flipping control **327**.
(37) The flash control **321** may be configured to turn on or off a flash.
(38) The setting control **322** may be configured to: adjust a video recording parameter (for example, resolution), enable or disable some video recording manners (for example, mute shooting), and the like.

(39) The preview box **323** may be configured to display an image collected by the camera **193** in real time. The first electronic device **100** may refresh display content in the first electronic device **100** in real time, so that a user previews an image currently collected by the camera **193**.
(40) One or more shooting mode options may be displayed in the camera mode option **324**. The one or more shooting mode options may include a portrait mode option, a photographing mode option, a video recording mode option, a professional mode option, and a more option. The one or more shooting mode options may be represented as text information on an interface, for example, “portrait”, “photographing”, “video recording”, “professional”, or “more”. This is not limited thereto. The one or more image shooting options may alternatively be represented as icons or interactive elements (interactive element, IE) in other forms on the interface. When detecting a user operation performed on the shooting mode option, the first electronic device **100** may enable a shooting mode selected by the user. In particular, when detecting a user operation performed on the more option, the first electronic device **100** may further display more other shooting mode options such as a slow-motion photographing mode option, and may present more photographing functions to the user. Not limited to that shown in FIG. **3**B, more options may not be displayed in the camera mode options, and the user may browse other shooting mode options by sliding left/right in the camera mode option **324**.
(41) The gallery shortcut button **325** may be used to start a gallery application. In response to a user operation performed on the gallery shortcut button **325**, for example, a tap operation, the first electronic device **100** may start the gallery application. In this way, the user can conveniently view a photographed photo and video without exiting the camera application and then starting the gallery application. The gallery application is an image management application on an electronic device such as a smartphone or a tablet computer, and may also be referred to as “Album”. A name of the application is not limited in this embodiment. The gallery application may support the user in performing various operations on a picture stored on the first electronic device **100**, for example, browsing, editing, deleting, and selecting.
(42) The video recording start control **326** may be used to listen to a user operation that triggers starting of recording. For example, as shown in FIG. **3**B, the video recording mode option is in a selected state. In response to a user operation performed on the video recording start control **326**, the first electronic device **100** may start video recording. The first electronic device **100** may save images collected by the camera **193** in a time period from starting video recording to ending video recording in a sequence of collection time as a video. Audio in the video may come from audio collected by an audio input apparatus (for example, a microphone) of the first electronic device **100** and/or audio collected by the audio input apparatus (for example, a microphone) of the TWS headset that establishes the Bluetooth connection to the first electronic device **100**.
(43) The camera flipping control **327** may be used to listen to a user operation that triggers camera flipping. In response to a user operation performed on the camera flipping control **327**, the first electronic device **100** may flip a camera. For example, the rear camera is switched to the front camera. In this case, the preview box **323** may display an image collected by the front camera shown in FIG. **3**B.
(44) The user interface **320** may further include more or less content. This is not limited in this embodiment of this application.
(45) In response to a user operation performed on the setting control **322**, the first electronic device **100** may display a user interface **330** shown in FIG. **3**C. The user interface **330** may include a return control **331**, a resolution control **332**, a geographical location control **333**, a shooting mute control **334**, and a sound source enhancement control **335**. The sound source enhancement control **335** may be used by the user to enable or disable a sound source enhancement function. When the sound source enhancement function is enabled, the first electronic device **100** turns on the microphone of the device when recording a video.
(46) As shown in FIG. **3**C, the sound source enhancement function is in a disabled state. In

response to the user operation performed on the sound source enhancement function, the first electronic device **100** may display a user interface **330** shown in FIG. **3**D. The user interface **330** may include a prompt box **336**. The prompt box **336** may include a text prompt "The microphone of the device is turned on when a video is recorded." The prompt box **336** may be used to prompt the user that when the sound source enhancement function is enabled, the first electronic device **100** turns on the microphone of the device in a video recording process. The prompt box **336** may further include a determining control **336**A. In response to a user operation performed on the determining control **336**A, the first electronic device **100** may display a user interface **330** shown in FIG. **3**E. In this case, the sound source enhancement function is an enabled state.
(47) It may be understood that if the first electronic device **100** is not connected to the TWS headset, the sound source enhancement function fails. That is, the user cannot use the foregoing sound source enhancement function to improve sound quality in the recorded video.
(48) In response to a user operation performed on the return control **331** shown in FIG. **3**E, the first electronic device **100** may display a user interface **320** shown in FIG. **3**F. In response to a user operation performed on the video recording start control **326**, the first electronic device **100** may start video recording.
(49) As shown in FIG. **3**G, both the left-ear TWS earphone **201** and the right-ear TWS earphone **202** are connected to the first electronic device **100**. The user wears the left-ear TWS earphone **201** and the right-ear TWS earphone **202**, and performs video recording by using the front camera of the first electronic device **100**. An audio signal in the video comes from a first audio signal collected by the microphone of the first electronic device **100**, a second audio signal collected by a microphone of the left-ear TWS earphone **201**, and a third audio signal collected by a microphone of the right-ear TWS earphone **202**. The first electronic device **100** may process the first audio signal, the second audio signal, and the third audio signal, to enhance a target sound in the first audio signal, the second audio signal, and the third audio signal, and suppress ambient noise in the first audio signal, to improve sound quality in the recorded video.
(50) In some embodiments, when the TWS headset establishes a connection to the first electronic device **100**, and when the first electronic device **100** detects a user operation of starting video recording, the first electronic device **100** may directly turn on the microphone to collect a sound. That is, when wearing the TWS headset and using the first electronic device **100** to record a video, the user may not need to manually enable the sound source enhancement function. When the user wears the TWS headset and uses the first electronic device **100** to record a video, the first electronic device **100** may automatically implement the sound source enhancement function, to provide sound quality in the recorded video.
(51) It should be noted that, in a process in which the first electronic device **100** processes the first audio signal, the second audio signal, and the third audio signal, because a specific time is required for transmitting the second audio signal and the third audio signal to the first electronic device **100**, time delays of different time lengths exist between the first audio signal, the second audio signal, and the third audio signal. The first electronic device **100** may perform delay alignment on the first audio signal, the second audio signal, and the third audio signal, to obtain a joint microphone array signal.
(52) In addition, to enhance a target sound in the joint microphone array signal and suppress ambient noise sound in the joint microphone array signal, the first electronic device may perform spatial filtering on the joint microphone array signal. The foregoing spatial filtering requires location information of a first microphone array formed by the microphone of the first electronic device **100**, the microphone of the left-ear TWS earphone **201**, and the microphone of the right-ear TWS earphone **202**. Because the left-ear TWS earphone **201** and the right-ear TWS earphone **202** are respectively worn on a left ear and a right ear of the user, the first electronic device **100** may determine the location information of each microphone in the first microphone array by using a face image collected by the front camera and posture information of the first electronic device **100**.

(53) The foregoing implementation method for the first electronic device to perform delay alignment, determine location information of each microphone in the first microphone array, and perform spatial filtering on the joint microphone array signal is specifically described in subsequent embodiments.
(54) User operations mentioned above and in subsequent embodiments are not limited in this embodiment of this application. For example, the user may indicate, by touching a location of a control on a display, the first electronic device **100** to execute an instruction corresponding to the control (for example, enabling the sound source enhancement function or starting video recording).
(55) With reference to the sound collecting scenario in the foregoing embodiment, the following describes a first microphone array and a sound collecting system that are provided in embodiments of this application and that are used for sound collecting.
(56) As shown in FIG. **4**, the sound collecting system may include a first electronic device **100**, a left-ear TWS earphone **201**, and a right-ear TWS earphone **202**. The first microphone array may include a first microphone **211**, a second microphone **230**, and a third microphone **220**. The first microphone **211** is a microphone of the first electronic device **100**. The second microphone **230** is a microphone of the left-ear TWS earphone **201**. The third microphone **220** is a microphone of the right-ear TWS earphone **202**.
(57) The first electronic device **100** may further include a speaker **210**, a camera **213**, a posture sensor **214**, a digital-to-analog converter (digital to analog converter, DAC) **215**A, an analog to digital converter (analog to digital converter, ADC) **215**B, an ADC **215**C, and a digital signal processor (digital signal processor, DSP) **216**.
(58) The speaker **210** may be configured to convert an audio electrical signal into a sound signal. The first electronic device **100** may play a sound by using the speaker **210**.
(59) The camera **213** may be the front camera in the camera **193** in the foregoing embodiment. The camera **213** may be configured to capture a static image or a video.
(60) The posture sensor **214** may be configured to measure posture information of the first electronic device **100**. The posture sensor **214** may include a gyroscope sensor and an acceleration sensor.
(61) The DAC **215**A may be configured to convert a digital audio signal into an analog audio signal.
(62) The ADC **215**B may be configured to convert an analog audio signal into a digital audio signal.
(63) The ADC **215**C may be configured to convert an analog image signal into a digital image signal.
(64) The DSP **216** may be configured to process a digital signal, for example, a digital image signal and a digital audio signal. The DSP **216** may include a signal generator **216**A, a delay alignment module **216**B, a coordinate resolving module **216**C, and a spatial filtering module **216**D.
(65) The signal generator **216**A may be configured to generate an alignment signal. The alignment signal is a digital audio signal. Because a frequency band of an audio signal that can be heard by a human ear is generally 20 Hz (Hz) to 20000 Hz, the foregoing alignment signal may be an audio signal whose frequency is in a frequency band range greater than 20000 Hz. In this way, the alignment signal can avoid a frequency band that can be heard by a human ear, and avoid interference to a user. The alignment signal may be used by the first electronic device **100** to perform delay alignment on an audio signal collected by the microphone in the first microphone array. When the first electronic device **100** detects a user operation of starting video recording, the signal generator **216**A may generate the alignment signal, and send the alignment signal to the DAC **215**A.
(66) In this embodiment of this application, a frequency of the foregoing alignment signal is not limited, and the alignment signal may alternatively be an audio signal whose frequency is 20000 Hz or a frequency within a frequency band range below 20000 Hz.

(67) The DAC **215**A may convert the alignment signal into an analog signal, and send the analog signal to the speaker **210**. The speaker **210** may then emit an alignment sound. The alignment sound is a sound corresponding to the foregoing alignment signal. The first microphone **211**, the second microphone **230**, and the third microphone **220** all can hear the alignment sound.
(68) In addition, the signal generator **216**A may further send the alignment signal to the delay alignment module **216**B.
(69) The delay alignment module **216**B may use the alignment signal to perform delay alignment on a first audio signal collected by the first microphone **211**, a second audio signal collected by the second microphone **230**, and a third audio signal collected by the third microphone **220**, to obtain a joint microphone array signal. The joint microphone array signal is a matrix. A first row, a second row and a third row of the matrix may be respectively the first audio signal, the second audio signal, and the third audio signal that have undergone delay alignment processing.
(70) The coordinate resolving module **216**C may be configured to determine coordinates of each microphone in the first microphone array in a world coordinate system. The microphones in the first microphone array are located in different electronic devices. A location of each microphone in the first microphone array varies based on a distance and a posture of a user wearing a TWS headset relative to the first electronic device **100**.
(71) The coordinate resolving module **216**C may resolve the coordinates of each microphone in the first microphone array in the world coordinate system by using a face image collected by the front camera (that is, the camera **213**) and the posture information of the first electronic device **100**. The camera **213** may send the collected image to the ADC **215**C. The ADC **2150** may convert the image into a digital signal, to obtain an image signal shown in FIG. **4**. The ADC **215**C may send the image signal to the coordinate resolving module **216**C. In addition, the posture sensor **214**C may send the collected posture information of the first electronic device **100** to the coordinate resolving module **216**C. The coordinate resolving module **216**C may obtain coordinate information of the first microphone array based on the image signal and the posture information of the first electronic device **100**. The coordinate information of the first microphone array includes coordinates of each microphone in the first microphone array in the world coordinate system.
(72) The spatial filtering module **216**D may be configured to perform spatial filtering on the joint microphone array signal. Based on the coordinate information of the first microphone array, the spatial filtering module **216**D may enhance a target sound from the user in the joint microphone array signal, and suppress ambient noise. The spatial filtering module **216**D may output a result audio signal. The result audio signal is the joint microphone array signal on which spatial filtering is performed.
(73) In a possible implementation, the spatial filtering module **216**D may perform spatial filtering on the first audio signal, the second audio signal, and the third audio signal together after delay processing. Specifically, the spatial filtering module **216**D may use a plurality of rows of data in a same column in the joint microphone array as one group of data, and perform spatial filtering on each group of data. Further, the spatial filtering module may output a signal on which spatial filtering is performed. The signal is the result audio signal.
(74) In another possible implementation, the spatial filter **216**D may perform spatial filtering on the first audio signal, the second audio signal, and the third audio signal separately after delay alignment processing. Specifically, the spatial filter **216**D may perform spatial filtering on the first row (that is, the first audio signal on which delay alignment processing is performed) in the joint microphone array, to obtain a first channel of audio. The spatial filter **216**D may perform spatial filtering on the second row (that is, the second audio signal on which delay alignment processing is performed) in the joint microphone array, to obtain a second channel of audio. The spatial filter **216**D may perform spatial filtering on the third row (that is, the third audio signal on which delay alignment processing is performed) in the joint microphone array, to obtain a third channel of audio. The first electronic device **100** may combine the first channel of audio, the second channel

of audio, and the third channel of audio into one channel of audio, to obtain the result audio signal.
(75) The first electronic device **100** may mix the result audio signal and a video collected by the camera, and then store the mixed result audio signal and the video locally or upload the mixed result audio signal and the video to a cloud server.
(76) The left-ear TWS earphone **201** may further include an ADC **231**A. The ADC **231**A may be configured to convert an analog audio signal into a digital audio signal. For example, the second microphone **230** may send a heard sound (that is, the analog audio signal) to the ADC **231**A. The ADC **231**A may convert the sound into the second audio signal (that is, the digital audio signal).
(77) The sound heard by the second microphone **230** may include the alignment sound, the target sound from the user (for example, a user voice, or a sound of playing a musical instrument by the user), the ambient noise, and the like. The second audio signal may include the alignment signal, a target sound signal from the user, and an ambient noise signal. The left-ear TWS earphone **201** may send the second audio signal to the first electronic device **100** through Bluetooth.
(78) The right-ear TWS earphone **202** may further include an ADC **221**A. The ADC **221**A may be configured to convert an analog audio signal into a digital audio signal. For example, the third microphone **220** may send a heard sound (that is, an analog audio signal) to the ADC **221**A. The ADC **221**A may convert the foregoing sound into a third audio signal (that is, the digital audio signal). The sound heard by the third microphone **220** may include the alignment sound, the target sound from the user (for example, a user voice or a sound of playing a musical instrument by the user), the ambient noise, and the like. The third audio signal may include the alignment signal, the target sound signal from the user, and the ambient noise signal. The right-ear TWS earphone **202** may send the third audio signal to the first electronic device **100** through Bluetooth.
(79) It can be learned from a location of each microphone in the first microphone array that a near field area formed by the first microphone array includes an area in which a user wearing a TWS headset and the first electronic device **100** are located. Compared with a microphone array in a single TWS earphone, the first microphone array has a larger size and a stronger spatial resolution capability; and can more accurately distinguish the target sound from the user in the near field area and the ambient noise from a far field area. In this way, the first electronic device **100** can better enhance the target sound and suppress the ambient noise when performing spatial filtering on the audio signal collected by the first microphone array, to improve sound quality in a recorded video.
(80) In addition, adding the microphone in the first electronic device **100** to the microphone array can reduce impact of a posture of wearing the TWS headset by the user on enhancing the target sound and reducing the ambient noise by the first electronic device **100**.
(81) The following specifically describes a method for performing delay alignment on an audio signal according to an embodiment of this application.
(82) Microphones in a first microphone array are distributed on different electronic devices. A first electronic device **100** may process an audio signal collected by the first microphone array, and mix the processed audio signal with a video collected by a camera. An audio signal collected by a microphone in a TWS headset needs to be sent to the first electronic device **100** through Bluetooth. The foregoing signal transmission introduces a delay difference of an order of hundreds of milliseconds. In addition, time points at which microphones in the first microphone array start to collect sound may be different, and time points at which the first electronic device **100** and the TWS headset process audio signals collected by respective microphones may also be different. The foregoing factors each cause a delay difference between a first audio signal, a second audio signal, and a third audio signal received by the first electronic device **100**.
(83) To resolve the foregoing delay difference problem, the first electronic device **100** may generate an alignment signal, and after converting the alignment signal into an analog audio sound effect, a speaker emits the alignment sound. The first electronic device may determine, by using the alignment signal, delay lengths between audio signals collected by the microphones.
(84) In some embodiments, when a user operation of starting video recording is detected, the first

electronic device **100** may emit the alignment sound. A first microphone, a second microphone, and a third microphone may respectively collect the first audio signal, the second audio signal, and the third audio signal. The first audio signal, the second audio signal, and the third audio signal each may include the alignment signal, a target sound signal from a user, and an ambient noise signal.
(85) As shown in FIG. **5**A, a delay alignment module **216**B of a first electronic device **100** may receive a first audio signal **502**, a second audio signal **503**, a third audio signal **504**, and an alignment signal **501** from a signal generator **216**A, Data that is in the alignment signal **501** and that corresponds to an alignment sound emitted at a time point T**0**, data that is in the first audio signal **502** and that is collected by a first microphone at the time point T**0**, data that is in the second audio signal **503** and that is collected by a second microphone at the time point T**0**, and data that is in the third audio signal **504** and that is collected by a third microphone at the time point T**0** are not aligned.
(86) To determine data that is collected at a same time point in the first audio signal **501**, the second audio signal **502**, and the third audio signal **503**, the delay alignment module **216**B may perform delay alignment on the first audio signal **502**, the second audio signal **503**, and the third audio signal **504** with the alignment signal **501**.
(87) Specifically, the delay alignment module **216**B may perform high-pass filtering on the first audio signal **501**, the second audio signal **502**, and the third audio signal **503**, to obtain alignment signals in these audio signals. The delay alignment module **216**B may perform delay correlation detection on the alignment signals in the audio signals and the alignment signal **501**, to obtain delay lengths between the alignment signals. The delay lengths between the alignment signals are the delay lengths between the first audio signal **502**, the second audio signal **503**, the third audio signal **504**, and the alignment signal **501**.
(88) The delay alignment module **216**B may perform delay processing of different time lengths on the alignment signals in the first audio signal **501**, the second audio signal **502**, and the third audio signal **503**. Then, the delay alignment module **216**B may compare correlation between the alignment signals that undergoes delay processing and the alignment signal **501**.
(89) As shown in FIG. **5**B, the delay alignment module **216**B may perform high-pass filtering on the first audio signal **502**, to obtain a signal **512** including the alignment signal. The delay alignment module **216**B may compare a correlation between the alignment signal **501** and the signal **512** starting at a different time point. The alignment signal **501** may be, for example, an audio signal with a time length of one second. The delay alignment module **216**B may determine that a correlation between the alignment signal **501** and a part of the signal **512** starting, from a time point Δt**1** and with a time length of one second is highest. In this case, the delay alignment module may determine that a delay length between the first audio signal **502** and the alignment signal **501** is Δt**1**.
(90) The delay alignment module **216**B may perform high-pass filtering on the second audio signal **503**, to obtain a signal **513** including the alignment signal. The delay alignment module **216**B may compare a correlation between the alignment signal **501** and the signal **513** starting at a different time point. The alignment signal **501** may be, for example, an audio signal with a time length of one second. The delay alignment module **216**B may determine that a correlation between the alignment signal **501** and a part of the signal **513** starting from a time point Δt**2** and with a time length of one second is highest. In this case, the delay alignment module may determine that a delay length between the second audio signal **503** and the alignment signal **501** is Δt**2**.
(91) The delay alignment module **216**B may perform high-pass filtering on the third audio signal **504**, to obtain a signal **514** including the alignment signal. The delay alignment module **216**B may compare a correlation between the alignment signal **501** and the signal **514** starting at a different time point. The alignment signal **501** may be, for example, an audio signal with a time length of one second. The delay alignment module **216**B may determine that a correlation between the alignment signal **501** and a part of the signal **514** starting from a time point Δt**3** and with a time

length of one second is highest. In this case, the delay alignment module may determine that a delay length between the third audio signal **504** and the alignment signal **501** is Δt**3**.

(92) A time length of the alignment signal **501** is not limited in this embodiment of this application. The alignment signal **501** may be an entire alignment signal generated by the signal generator **216**A. Optionally, the alignment signal **501** may be a part of the entire alignment signal generated by the signal generator **216**B. For example, the alignment signal generated by the signal generator **216**B is a periodic signal. The signal generator **216**B may generate alignment signals of a plurality of periods. The delay alignment module **216**B may perform delay correlation detection on the signal **512**, the signal **513**, and the signal **514** by using an alignment signal of one period, to determine delay lengths between the first audio signal, the second audio signal, and the third audio signal.

(93) The foregoing delay correlation detection may be a correlation detection method such as a cross-correlation time estimation method. A method for performing delay correlation detection by the delay alignment module **216**B is not limited in this embodiment of this application.

(94) When the delay lengths are determined, the delay alignment module **216**B may determine, by subtracting the delay length corresponding to an audio signal front a time point at which the audio signal is received, data that is collected at a same time point by the microphones belonging to the first microphone array and that is in the audio signals. The delay alignment module **216**B may output a joint microphone array signal. The joint microphone array signal may include a plurality of audio signals. The plurality of audio signals are audio signals obtained after the alignment signals in the first audio signal **502**, the second audio signal **503**, and the third audio signal **504** are filtered out. In the plurality of audio signals included in the joint microphone array signal, data corresponding to a same time point is data collected by the microphones in the first microphone array at the same time point. In this way, the delay alignment module **216**B may eliminate a delay difference caused by signal transmission.

(95) Examples in FIG. **5**A and FIG. **5**B are merely used to explain this application and shall not constitute a limitation.

(96) In some embodiments, the delay alignment module **216**B may further use any one of the first audio signal **502**, the second audio signal **503**, and the third audio signal **504** as a reference audio signal, and perform delay alignment between another audio signal and the reference audio signal. For example, the delay alignment module **216**B may use the first audio signal **502** as a reference audio signal. The delay alignment module **216**B may perform high-pass filtering on the first audio signal **502**, the second audio signal **503**, and the third audio signal **504**, to obtain alignment signal parts in the audio signals. The delay alignment module **216**B may perform delay processing of different time lengths on the alignment signal parts in the second audio signal **503** and the third audio signal **504**, to determine delay lengths at which a correlation between the alignment signal parts of the second audio signal **503** and the third audio signal **504** and the alignment signal part of the first audio signal **501** is the highest. In this way, the delay alignment module **216**B may determine the delay lengths of the second audio signal **503** and the third audio signal **504** relative to the first audio signal **502**.

(97) In some embodiments, the first electronic device **100** may generate the alignment signal in a preset time period (for example, a preset time period before video recording starts or a preset time period after video recording starts) for starting video recording, and emit the alignment sound. The first electronic device **100** may determine the delay lengths between the first audio signal, the second audio signal, and the third audio signal based on the alignment signal of the preset time period. The delay lengths generally do not change or change very little in a video recording process of the first electronic device **100**. In a video recording process after the preset time period, the first electronic device **100** may perform delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the delay lengths. In addition, the first electronic device **100** may stop generating the alignment signal, to save power consumption of the first electronic

device **100**.
(98) The following specifically describes a method for determining coordinate information of a first microphone array according to an embodiment of this application.
(99) A first electronic device **100** may perform spatial filtering on the foregoing joint microphone array signal, to enhance a target sound from a user and suppress ambient noise. In the foregoing spatial filtering process, the first electronic device **100** needs to determine a direction of a target sound signal and a direction of the ambient noise by determining the coordinate information of the first microphone array.
(100) Microphones in the first microphone array are separately distributed in a TWS headset and the first electronic device **100**. The foregoing TWS headset is usually worn on a left ear and a right ear of the user. The first electronic device **100** may determine location information of the first microphone array by using a face image collected by a front camera and posture information of the first electronic device **100**.
(101) 1. The first electronic device **100** may determine coordinates of each microphone in the first microphone array in a first electronic device coordinate system.
(102) The first electronic device **100** may first determine a conversion relationship between a three-dimensional (3 dimensions, 3D) head coordinate system and the first electronic device coordinate system based on a face image. Then, the first electronic device **100** may determine coordinates of the microphone of the TWS headset in the first microphone array in the first electronic device coordinate system by converting coordinates of outer auricles of a left ear and a right ear of a user in the 3D head coordinate system into coordinates in the first electronic device coordinate system.
(103) The 3D head coordinate system may be determined based on a standard head model. Each key point on a face may correspond to determined coordinates in the 3D head coordinate system. As shown in FIG. **6**A, a 3D head coordinate system x.sub.h-y.sub.h-z.sub.h may be a three-dimensional coordinate system established by using a location of a nose tip in a standard head as an origin, using a direction perpendicular to a face as a direction of an x-axis, using a horizontal direction parallel to the face as a direction of a y-axis, and using a vertical direction parallel to the face as a direction of a z-axis. The first electronic device **100** may store related data of the 3D head coordinate system x.sub.h-y.sub.h-z.sub.h. A method for establishing the 3D head coordinate system is not limited in this embodiment of this application.
(104) The first electronic device **100** may determine an extrinsic parameter matrix based on a relationship between coordinates of a key point of a face image in a pixel coordinate system and coordinates of a corresponding key point in a 3D head coordinate system. The pixel coordinate system is a two-dimensional coordinate system, and may be used to reflect an arrangement status of pixels in an image. For example, the pixel coordinate system may be a two-dimensional coordinate system established by using any pixel of the image as an origin and using directions parallel to two sides of an image plane as directions of an x-axis and a y-axis. The extrinsic parameter matrix may be used to describe a conversion relationship between the 3D head coordinate system x.sub.h-y.sub.hz.sub.h and the first electronic device coordinate system x.sub.d-y.sub.d-z.sub.d.
(105) Specifically, the face image collected by the front camera of the first electronic device **100** may include N key points. The N key points are respectively p.sub.1, p.sub.2, . . . , p.sub.i, . . . , p.sub.N, where i is a positive integer less than or equal to N. The N key points may be any N key points in an area in which the face is located in the face image, for example, a key point in an area in which a forehead area is located, a key point in an area in which a cheek is located, and a key point in an area in which a lip is located. A quantity of the foregoing key points is not limited in this embodiment of this application.
(106) The first electronic device **100** may determine coordinates {h.sub.i=[x.sub.i, y.sub.i, z.sub.i].sup.T, 2, . . . N} of the N key points in the 3D head coordinate system x.sub.h-y.sub.h-z.sub.h. The first electronic device **100** may determine coordinates {g.sub.i=[u.sub.i, v.sub.i].sup.T, i=1,2, . . . , N} of the N key points in the pixel coordinate system by using a key point detection

algorithm. The key point detection algorithm may be, for example, a method for determining key point coordinates by using a trained neural network model. The key point detection algorithm is not limited in this embodiment of this application.
(107) The coordinates of the N key points in the pixel coordinate system and the coordinates of the N key points in the 3D head coordinate system may have a relationship in the following formula (1):
g.sub.i=*C**(*R***h*.sub.i+*T*) (1) where C is an intrinsic parameter matrix. The intrinsic parameter matrix may be used to describe a conversion relationship between the pixel coordinate system and the first electronic device coordinate system x.sub.d-y.sub.d-z.sub.d. The intrinsic parameter matrix is related only to a parameter of a camera of the first electronic device **100**. The intrinsic parameter matrix may be obtained by using a camera calibration method. For a specific implementation method for the camera calibration, refer to a camera calibration method in the conventional technology. Details are not described herein. The first electronic device **100** may store the intrinsic parameter matrix C. R is a rotation matrix. T is a center offset vector. The foregoing R and T jointly form an extrinsic parameter matrix [R|T].
(108) The first electronic device **100** may resolve the extrinsic parameter matrix based on the coordinates of the N key points in the pixel coordinate system and the coordinates of the N key points in the 3D head coordinate system.

(109) $$R(\ , \ , \ ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos & \sin \\ 0 & -\sin & \cos \end{bmatrix} \begin{bmatrix} \cos & 0 & \sin \\ 0 & 1 & 0 \\ -\sin & 0 & \cos \end{bmatrix} \begin{bmatrix} \cos & -\sin & 0 \\ \sin & \cos & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$$T = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (3)$$

(110) As shown in FIG. **6**B, α, β, γ may be deflection angles between coordinate axes of the 3D head coordinate system x.sub.h-y.sub.h-z.sub.h and the first electronic device coordinate system x.sub.d-y.sub.d-z.sub.d, and x, y, and z in the center offset vector T may be offsets on an x.sub.d axis, a y.sub.d axis, and a z.sub.d axis in the first electronic device coordinate system when the origin of the 3D head coordinate system is offset to an origin of the first electronic device coordinate system.
(111) The first electronic device **100** may store the extrinsic parameter matrix.
(112) Further, the first electronic device **100** may determine coordinates of key points at which the outer auricles of the left-ear and the right-ear are located in the 3D head coordinate system. The coordinates of the outer auricle of the left-ear in the 3D head coordinate system may be h.sub.L=[x.sub.L, y.sub.L, z.sub.L].sup.T. The coordinates of the outer auricle of the right-ear in the 3D head coordinate system may be h.sub.R=[x.sub.R, y.sub.R, z.sub.R].sup.T. The first electronic device **100** may use h.sub.L and h.sub.R as coordinates of a second microphone **230** of a left-ear TWS earphone **201** and a third microphone **220** of a right-ear TWS earphone **202** in the 3D head coordinate system respectively.
(113) Because the extrinsic parameter matrix may be used to describe the conversion relationship between the 3D head coordinate system and the first electronic device coordinate system, the first electronic device **100** may convert the coordinates of the second microphone **230** and the third microphone **220** in the 3D head coordinate system into coordinates in the first electronic device coordinate system. The coordinates of the second microphone **230** in the first electronic device coordinate system may be e.sub.2=h.sub.L*[R|T]. The coordinates of the third microphone **220** in the first electronic device coordinate system may be e.sub.3=h.sub.R*[R|T].
(114) In addition, the first electronic device **100** may store coordinates e.sub.1 of the first microphone **211** in the foregoing embodiment in the first electronic device coordinate system.

(115) In this way, the first electronic device **100** may determine coordinates E={e.sub.1, e.sub.2, e.sub.3} of each microphone in the first microphone array in the first electronic device coordinate system.
(116) 2: The first electronic device **100** may determine a conversion relationship between the first electronic device coordinate system and a world coordinate system, and resolve coordinates of each microphone in the first microphone array in the world coordinate system.
(117) A placement posture of the first electronic device **100** is unknown and time-varying. The target sound that needs to be collected by the first microphone array includes the voice of the user, and may include the sound of playing a musical instrument by the user. In a spatial filtering process, using the coordinates of each microphone in the first microphone array in the first electronic device coordinate system by the first electronic device reduces an effect of spatial filtering. To better improve an effect of enhancing a target sound and reducing ambient noise, the first electronic device **100** may convert the coordinates of each microphone in the first microphone array in the first electronic device coordinate system into the coordinates in the world coordinate system.
(118) Specifically, the first electronic device **100** may obtain posture information of the first electronic device **100** by using a posture sensor **214**. As shown in FIG. **6**C, a posture signal may include deflection angles d, β′, and γ′ between coordinate axes of the first electronic device coordinate system x.sub.d-y.sub.d-z.sub.d and the world coordinate system x.sub.w-y.sub.w-z.sub.w. The first electronic device **100** may determine coordinates E′={e.sub.1′, e.sub.2′, e.sub.3′} of each microphone in the first microphone array in the world coordinate system by using the posture information, where E′ and E may have a relationship shown in the following formula (4):

(119) $$E' = \{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos ' & \sin ' \\ 0 & -\sin ' & \cos ' \end{bmatrix}\begin{bmatrix} \cos ' & -\sin ' & 0 \\ \sin ' & \cos ' & 0 \\ 0 & 0 & 1 \end{bmatrix}\}^{-1} E \quad (4)$$

(120) In some embodiments, the process of determining the coordinate information of the first microphone array may be completed by the coordinate resolving module **216**C shown in FIG. **4**.
(121) The following specifically describes a spatial filtering method according to an embodiment of this application.
(122) When the joint microphone array signal and the coordinate information of the first microphone array are obtained, a first electronic device **100** may perform spatial filtering on the joint microphone array signal. The first electronic device **100** may first perform voice activity detection on the joint microphone array signal. The voice activity detection may be used to distinguish between a signal whose frequency point is on a target sound and a signal whose frequency point is on ambient noise in the joint microphone array signal. Further, the first electronic device **100** may update a noise spatial characteristic based on the signal whose frequency point is on the ambient noise. The first electronic device **100** may estimate a target steering vector based on the signal whose frequency point is on the target sound. Both the noise spatial characteristic and the target steering vector are parameters used to determine a parameter of a spatial filter. The noise spatial characteristic update may be used to reduce impact of the target sound on suppressing ambient noise by the spatial filter. The target steering vector estimation may be used to reduce impact of the ambient noise at a same frequency point as the target sound on suppressing ambient noise by the spatial filter, thereby improving an effect of spatial filtering.
(123) The following describes the foregoing methods for voice activity detection, noise spatial characteristic update, target steering vector estimation, and spatial filter determining.
(124) 1. Voice Activity Detection (Voice Activity Detection, VAD)
(125) In some embodiments, a sound collected by a first microphone array in a near field area mainly includes a user voice and a sound of playing a musical instrument by a user. That is, the

target sound signal that needs to be enhanced in the joint microphone array signal mainly includes a voice signal and a sound signal of a musical instrument. Based on a difference between a characteristic of a voice signal and a characteristic of a sound signal of a musical instrument and a characteristic of an ambient noise signal, a first electronic device **100** may perform voice activity detection on a joint microphone array signal by using a neural network model, to distinguish between a target sound signal whose frequency point is on a target sound and an ambient noise signal whose frequency point is on an ambient noise signal in the joint microphone array signal.
(126) The foregoing neural network model for performing voice activity detection may be obtained by training by using the voice signal and the sound signal of a musical instrument. The trained neural network model may be used to distinguish between the target sound signal and the ambient noise signal in the joint microphone array signal. For example, if a voice signal or a sound signal of a musical instrument whose frequency point is on a voice or a musical instrument sound is input into a trained neural network model, the trained neural network model may output a label 1. The label 1 may indicate that an input received by the trained neural network model is the target sound signal whose frequency point is on the target sound. If an ambient noise signal whose frequency point is on the ambient noise is input into the trained neural network model, the trained neural network model may output a label 0. The label 1 may indicate that an input received by the trained neural network model is not the target sound signal (for example, is an ambient noise signal).
(127) The first electronic device **100** may store the trained neural network model.
(128) A method fir training the neural network model is not limited in this embodiment of this application. For a training method of the neural network model, refer to a specific implementation method in the conventional technology. Details are not described herein again. The neural network model may be a convolutional neural network model, a deep neural network model, or the like. A type of the neural network model is not limited in this embodiment of this application.
(129) In some embodiments, the first electronic device **100** may use the joint microphone array signal and the coordinate information of the first microphone array as an input of the trained neural network model. In this way, the trained neural network model can adapt to impact of a change in the coordinate information of the first microphone array on voice activity detection. In a case in which the first microphone array has different structures, the trained neural network model can better distinguish between the target sound signal and the ambient noise signal in the joint microphone array signal.
(130) 2. Noise Spatial Characteristic Update
(131) The first electronic device **100** may update a noise spatial characteristic of each frequency point in the joint microphone array signal by using a detection result of the voice activity detection, to reduce interference of the target sound signal to noise spatial characteristic of ambient noise.
(132) In a possible implementation, the noise space characteristic may be expressed by using a noise covariance matrix. The first electronic device **100** may update the noise covariance matrix based on the detection result of the voice activity detection. Specifically, a short-time Fourier transform (short time fourier transform, STFT) of the joint microphone array signal at a time point t and a frequency point f is X.sub.t(f). A noise covariance matrix at a previous time point of the time point t is R.sub.t−1(f). The first electronic device **100** performs voice activity detection on the joint microphone array signal. A detection result of the joint microphone array signal at the frequency point f is vad(f). The first electronic device **100** may update the noise covariance matrix at the time point t according to the following formula (5):

(133) $$R_t(f) = \begin{cases} R_{t-1}(f) & \text{vad}(f) = 1 \\ (1 - \text{fac}) * R_{t-1}(f) + \text{fac} * X_t(f) * X_t^H(f) & \text{vad}(f) = 0 \end{cases}, \quad (5)$$

where R.sub.t(f) is the noise covariance matrix at the time point t. fac is a smoothing factor. fac is greater than or equal to 0 and less than or equal to 1. fac A value of may be preset based on experience. A specific value of fac is not limited in this embodiment of this application.

X.sub.t.sup.H(f) is a conjugate transposition of X.sub.t(f). vad(f)−1 may indicate that an output label of the neural network model used for voice activity detection is 1. That is, a signal of the joint microphone array signal at the frequency point f is the target sound signal. vad(f)=0 may indicate that an output label of the neural network model used for voice activity detection is 0. That is, a signal of the joint microphone array signal at the frequency point f is an ambient noise signal.
(134) It can be learned from the foregoing method for updating a noise spatial characteristic that the first electronic device **100** may smoothly update a noise control characteristic based on whether the joint microphone array signal of the frequency point f is the target sound signal or the ambient noise sound signal. This can reduce a sudden change of ambient noise and impact of the target sound on suppressing ambient noise by the spatial filter.
(135) In addition to the foregoing method for updating the noise spatial characteristic, the first electronic device **100** may further update the noise spatial characteristic by using another method.
(136) 3. Target Steering Vector Estimation
(137) The first electronic device **100** may estimate the target steering vector based on the coordinate information of the first microphone array and a sound propagation model. The target steering vector may be used to represent a direction of the target sound signal. In a possible implementation, the target steering vector may be determined by using different delay times of arrival of the target sound signal at each microphone in the first microphone array. For the foregoing target steering vector estimation method, refer to a method for estimating a target steering vector in an existing spatial filtering technology. Details are not described herein again.
(138) In some embodiments, the first electronic device **100** may further use a subspace projection method to improve precision of the target steering vector. Improving precision of the target steering vector helps the first electronic device **100** more accurately distinguish between the direction of the target sound signal and a direction of the ambient noise signal.
(139) The target steering vector estimation may be used to reduce impact of the ambient noise at a same frequency point as the target sound on suppressing ambient noise by the spatial filter, thereby improving an effect of spatial filtering.
(140) 4. Spatial Filter Determining
(141) A spatial filter may be used to process a plurality of microphone signals (that is, the joint microphone array signal), suppress a signal in a non-target direction (that is, the ambient noise signal), and enhance a signal in a target direction (that is, the target sound signal). The first electronic device **100** may determine the spatial filter by using a method such as a minimum variance distortionless response (minimum variance distortionless response, MVDR) beamforming algorithm, a linearly constrained minimum variance (linearly constrained minimum variance, LCMV) beamforming algorithm, or a generalized sidelobe canceller (generalized sidelobe canceller, GSC). A specific method for determining the spatial filter is not limited in this embodiment of this application.
(142) For example, the first electronic device **100** may determine a control filter by using the MVDR beamforming algorithm. A principle of the method is to select a proper filter parameter under a constraint condition that a desired signal has no distortion, to minimize average power output by the joint microphone array signal. The first electronic device **100** may determine an optimal spatial filter by using the noise spatial characteristic and the target steering vector. The optimal spatial filter can minimize impact of the ambient noise signal under a constraint condition that the target steering vector passes through without distortion. The spatial filter may be designed according to the following formula (6):
(143) $$w(f) = \frac{R_t^{-1}(f) * a_t(f)}{a_t^H(f) * R_t^{-1}(f) * a_t(f)}, \quad (6)$$
where w(f) is an optimal filtering weight coefficient of the spatial filter, and R.sub.t(f) and a, (f) are respectively a noise covariance matrix and a target steering vector of a time point t and a frequency point f.

(144) When the foregoing optimal spatial filter is obtained, the first electronic device **100** may input the joint microphone array signal into the optimal spatial filter for spatial filtering. The first electronic device **100** may obtain a result audio signal by using the spatial filter. The result audio signal is an audio signal obtained after the target sound signal in the joint microphone array signal is enhanced and the ambient noise in the joint microphone array signal is suppressed.
(145) In some embodiments, the foregoing spatial filtering process may be completed by the spatial filtering module **216**D shown in FIG. **4**,
(146) The following describes a method for turning on a first microphone array according to an embodiment of this application.
(147) The first microphone array may include a first microphone in a first electronic device **100**, a second microphone in a left-ear TWS earphone **201**, and a third microphone in a right-ear TWS earphone **202**. Before the first microphone array performs sound collecting, each of the left-ear TWS earphone **201** and the right-ear TWS earphone **202** establishes a Bluetooth connection to the first electronic device **100**. Not limited to connection through Bluetooth, the TWS headset may alternatively establish a communication connection to the first electronic device **100** in another communication manner.
(148) FIG. **7**A is an example of a flowchart of a method for turning on a first microphone array. As shown in FIG. **7**A, the method may include steps S**101** to S**110**.
(149) S**101**: A first electronic device **100** receives a user operation of starting video recording.
(150) A user may use the first electronic device **100** to perform video recording in a scenario in which a front camera is used to record a video, such as live broadcast or vlog shooting. The first electronic device **100** may receive the user operation of starting video recording. The user operation of starting video recording may be, for example, a user operation performed on the video recording start control **326** as shown in FIG. **3**F.
(151) Before the user operation of starting video recording is received, the front camera of the first electronic device **100** is in an enabled state. The first electronic device **100** may display, in the preview box **323** shown in FIG. **3**F, an image collected by the front camera in real time.
(152) S**102**: A left-ear TWS earphone **201** performs wearing detection.
(153) S**103**: A right-ear TWS earphone **202** performs wearing detection.
(154) In some embodiments, each of the left-ear TWS earphone **201** and the right-ear TWS earphone **202** may include a wearing detection module. The wearing detection module may be configured to detect whether a user wears the TWS headset.
(155) The wearing detection module may include a temperature sensor. The TWS headset may obtain a surface temperature of an earpiece of the TWS headset by using an optical proximity sensor. When detecting that the surface temperature of the earpiece of the TWS headset exceeds a preset value, the TWS headset may determine that the user wears the TWS headset. In this way, the TWS headset can implement wearing detection by using the temperature sensor. When detecting the TWS headset detects that the user wears the TWS headset, the TWS headset may wake up a main processor, to implement functions such as playing music and collecting a sound. When detecting that the user does not wear the TWS headset, the TWS headset may control the main processor and components such as a speaker and a microphone to be in a sleep state. This can reduce power consumption of the TWS headset.
(156) In addition to the foregoing method for performing wearing detection by using the temperature sensor, the TWS headset may alternatively perform wearing detection by using an optical proximity sensor, a motion sensor, a pressure sensor, or the like. A method for performing wearing detection by the TWS headset is not limited in this embodiment of this application.
(157) S**104**: The first electronic device TOO determines that the user wears the right-ear TWS earphone **202**.
(158) S**105**: The first electronic device **100** determines that a user wears the left-ear TWS earphone **201**.

(159) When receiving the user operation of starting video recording in the foregoing step S**101**, the first electronic device **100** may send a message to the left-ear TWS earphone **201** and the right-ear TWS earphone **202**, to query a wearing detection result. The left-ear TWS earphone **201** and the right-ear TWS earphone **202** may send the wearing detection result to the first electronic device **100**. The first electronic device **100** may determine, based on the received wearing detection result, that the user wears the left-ear TWS earphone **201** and the right-ear TWS earphone **202**.
(160) An execution sequence of step S**104** and step S**105** is not limited in this embodiment of this application.
(161) S**106**: The first electronic device **100** enables a first microphone.
(162) When determining that the user wears the left-ear TWS earphone **201** and the right-ear TWS earphone **202**, the first electronic device **100** may enable the first microphone to perform sound collecting.
(163) S**107**: The first electronic device **100** sends a microphone turn-on instruction to the right-ear TWS earphone **202**.
(164) S**108**: The first electronic device **100** sends a microphone turn-on instruction to the left-ear TWS earphone **201**.
(165) When determining that the user wears the left-ear TWS earphone **201**, the first electronic device **100** may further send the microphone turn-on instruction to the left-ear TWS earphone **201**.
(166) When determining that the user wears the right-ear TWS earphone **202**, the first electronic device **100** may further send the microphone turn-on instruction to the right-ear TWS earphone **202**.
(167) An execution sequence of step S**106**, step S**107**, and step S**108** is not limited in this embodiment of this application.
(168) S**109**: The left-ear TWS earphone **201** turns on a second microphone.
(169) When the wearing detection result indicates that the user wears the left-ear TWS earphone **201**, and the left-ear TWS earphone **201** has received the microphone turn-on instruction from the first electronic device **100**, the left-ear TWS earphone **201** may turn on the second microphone.
(170) S**110**: The right-ear TWS earphone **202** turns on a third microphone.
(171) When the wearing detection result indicates that the user wears the right-ear TWS earphone **202**, and the right-ear TWS earphone **202** has received the microphone turn-on instruction from the first electronic device **100**, the right-ear TWS earphone **202** may turn on the third microphone.
(172) That is, when the user wears the left-ear TWS earphone **201** and the right-ear TWS earphone **202** and performs video recording by using the first electronic device **100**, microphones configured to perform sound collecting may include the first microphone, the second microphone, and the third microphone. That is, the first microphone is turned on. This is not limited to the foregoing video recording scenario. A scenario in which the first microphone is turned on may alternatively be a scenario in which the user wears the left-ear TWS earphone **201** and the right-ear TWS earphone **202** and performs a video call by using the first electronic device **100**, or the like.
(173) In some embodiments, the left-ear TWS earphone **201** may turn on the second microphone when performing wearing detection and determining that the left-ear TWS earphone **201** is in an in-ear state. The right-ear TWS earphone **202** may turn on the third microphone when performing wearing detection and determining that the right-ear TWS earphone **202** is in the in-ear state. That is, the first electronic device **100** may not need to send the microphone turn-on instruction to the left-ear TWS earphone **201** and the right-ear TWS earphone **202**.
(174) Based on the method for turning on the first microphone array shown in FIG. **7**A, the following describes a sound collecting method provided in an embodiment of this application.
(175) FIG. **7**B shows an example of a flowchart of a sound collecting method. As shown in FIG. **7**B, the method may include steps S**201** to S**210**. Steps S**201** to S**207** are a process of performing sound collecting by a first microphone array. Steps S**208** to S**210** are a process in which a first electronic device **100** processes an audio signal collected by the first microphone array.

(176) 1. (S**201** to S**207**) The first microphone array performs sound collecting.
(177) S**201**: The first electronic device **100** receives a user operation of starting video recording.
(178) For the foregoing user operation of starting hiring, refer to the description of step S**101** in FIG. **7**A, Details are not described herein again.
(179) S**202**: The first electronic device **100** emits an alignment sound.
(180) For the alignment signal, refer to the description in the foregoing embodiments. Details are not described herein again.
(181) S**203**: A left-ear TWS earphone **201** collects a second audio signal by using a second microphone, where the second audio signal includes an alignment signal, a target sound signal from a user, and an ambient noise signal.
(182) S**204**: A right-ear TWS earphone **202** collects a third audio signal by using a third microphone, where the third audio signal includes an alignment signal, a target sound signal from the user, and an ambient noise signal.
(183) S**205**: The first electronic device **100** collects a first audio signal by using a first microphone, where the first audio signal includes an alignment signal, a target sound signal from the user, and an ambient noise signal.
(184) In a process in which the first electronic device **100** turns on a front camera to record a video, sounds near the first electronic device **100**, the left-ear TWS earphone **201**, and the right-ear TWS earphone **202** may include an alignment sound, a user voice, a sound of playing a musical instrument by the user, and ambient noise. The target sound collected by the first microphone array includes the user voice and the sound of playing the musical instrument by the user. The target sound is a sound that is reserved and that is clearly recorded in the video that is expected to be recorded. The ambient noise is a sound that is not expected to be recorded.
(185) All of the first microphone, the second microphone, and the third microphone may collect sounds near the first electronic device **100**, the left-ear TWS earphone **201**, and the right-ear TWS earphone **202**. The first microphone, the second microphone, and the third microphone may send the sound collected by the first microphone, the second microphone, and the third microphone to a related audio processing module (for example, an ADC) for processing, and respectively obtain the first audio signal, the second audio signal, and the third audio signal.
(186) S**206**: The left-ear TWS earphone **201** may send the second audio signal to the first electronic device **100** through Bluetooth.
(187) S**207**: The right-ear TWS earphone **202** may send the third audio signal to the first electronic device **100** through Bluetooth.
(188) 2. (S**208** to S**210**) The first electronic device **100** processes the audio signal collected by the first microphone array.
(189) S**208**: The first electronic device **100** performs delay alignment on the first audio signal, the second audio signal, and the third audio signal, to obtain a joint microphone array signal.
(190) S**209**: The first electronic device **100** determines coordinate information of the first microphone array with reference to a face image collected by a front camera and posture information of the first electronic device **100**.
(191) S**210**: The first electronic device **100** performs, based on the coordinate information of the first microphone array, spatial filtering on the joint microphone array signal by using a spatial filter.
(192) When obtaining a result audio signal obtained after spatial filtering, the first electronic device **100** may mix the result audio signal with a video captured by a camera in a process from starting video recording to ending video recording. The first electronic device **100** may determine a delay length of the foregoing result audio signal and the video based on a time point at which any one or more microphones in the first microphone array start to perform sound collecting and a time point at which the camera of the first electronic device **100** starts to perform image collecting. According to the delay length of the result audio signal and the video, the first electronic device **100** may ensure that a result audio signal and the video are aligned in time when mixing the result audio

signal and the video. A method for performing delay alignment processing on the result audio signal and the video by the first electronic device **100** is not limited in this embodiment of this application.
(193) Further, the first electronic device **100** may locally store mixed audio and video data or upload the mixed audio and video data to a cloud server.
(194) For implementation methods of steps S**208** to S**210**, refer to the description in the foregoing embodiment. Details are not described herein again.
(195) It should be noted that, in some embodiments, the first electronic device **100** may determine the coordinate information of the first microphone array by using a face image collected within a preset time period after video recording starts and the posture information of the first electronic device **100**. The first electronic device **100** may store the coordinate information of the first microphone array, and use the coordinate information of the first microphone array when performing spatial filtering on the joint microphone array signal in the video recording process. That is, the first electronic device **100** may not repeatedly measure the coordinate information of the first microphone array in one video recording process. This can reduce power consumption of the first electronic device **100**. In addition, in one video recording process, a distance and a direction between the user wearing the TWS headset and the first electronic device **100** generally do not change greatly. That the first electronic device **100** uses the coordinate information of the first microphone array determined in the preset time period after the video recording starts as the coordinate information of the first microphone array in the video recording process has little impact on an effect of enhancing the target sound and suppressing the ambient noise.
(196) In some other embodiments, the first electronic device **100** may determine the coordinate information of the first microphone array by using a face image collected within a preset time period after video recording starts and the posture information of the first electronic device **100**. In a subsequent phase of the video recording, the first electronic device **100** may determine a distance and a direction between the user and the first electronic device **100** at intervals of a fixed time period. If the first electronic device **100** determines that a variation of the distance and the direction between the user and the first electronic device **100** exceeds a preset variation, the first electronic device **100** may re-determine the coordinate information of the first microphone array based on the face image currently collected by the front camera and the posture information of the first electronic device **100**. Further, the first electronic device **100** may perform spatial filtering on the joint microphone array signal by using the re-determined coordinate information of the first microphone array. If the first electronic device **100** determines that the variation of the distance and the direction between the user and the first electronic device **100** does not exceed the preset variation, the first electronic device **100** may continue to perform spatial filtering on the joint microphone array signals by using the currently stored coordinate information of the first microphone array. The foregoing method reduces a quantity of times that the first electronic device **100** determines the coordinate information of the first microphone array reduces power consumption of the first electronic device **100**, and reduces impact of a change of the coordinate information of the first microphone array on spatial filtering in a video recording process.
(197) The first electronic device **100** may determine the variation of the distance and the direction between the user and the first electronic device **100** by detecting a change of a size and a location of a face box in an image collected by the front camera. Optionally, the first electronic device **100** may further determine the variation of the distance and the direction between the user and the first electronic device **100** in a manner such as using an optical proximity sensor or sound wave ranging. This is not limited in this embodiment of this application.
(198) It can be learned from the sound collecting method shown in FIG. **7**B that, when recording a video in a case in which a communication connection is established between the first electronic device **100** and the TWS headset, the first electronic device **100** may still turn on the microphone of the first electronic device **100** to perform sound collecting. The microphone of the first electronic

device **100** and the microphone of the TWS headset may form the first microphone array. A near field area formed by the first microphone array includes an area in which a user wearing the TWS headset and the first electronic device **100** are located. Compared with a microphone array formed by only microphones of a TWS headset, the first microphone array has a larger size and a stronger spatial resolution capability, and can more accurately distinguish a target sound from a user in the near field area and ambient noise from a far field area. In this way, the first electronic device **100** can better enhance the target sound and suppress the ambient noise when performing spatial filtering on the audio signal collected by the first microphone array, to improve sound quality in a recorded video.
(199) In addition, adding the microphone in the first electronic device **100** to the microphone array can reduce impact of a posture of wearing the TWS headset by the user on enhancing the target sound and reducing the ambient noise by the first electronic device **100**.
(200) The sound collecting method provided in this application is particularly applicable to a scenario in which a user wearing a TWS headset performs video recording by using the first electronic device **100** for live broadcast, vlog shooting, or the like. In addition to the foregoing video recording scenario, the voice collection method provided in this application may be alternatively applied to another scenario such as a video call.
(201) In some embodiments, a microphone of one of the left-ear TWS earphone **201** and the right-ear TWS earphone **202** may form a microphone array with the microphone of the first electronic device **100**. The microphone array may also be applicable to the sound collecting method provided in embodiments of this application. For example, in a scenario in which the user wears the left-ear TWS earphone **201** and the right-ear TWS earphone **202** and uses the first electronic device **100** for live broadcast, the first microphone of the first electronic device **100** and the second microphone of the left-ear TWS earphone **201** may perform sound collecting. The first electronic device **100** may obtain the first audio signal by using the first microphone. The left-ear TWS earphone **201** may obtain the second audio signal by using the second microphone. The left-ear TWS earphone **201** may send the second audio signal to the first electronic device **100**. The first electronic device **100** may perform noise reduction processing on the first audio signal and the second audio signal, to obtain an audio signal in a live video.
(202) A near field area of the microphone array formed by the microphone of one earphone and the microphone of the first electronic device **100** may still include a location of a sound source of a target sound, such as a user voice or a sound of playing a musical instrument by the user. In addition, using the microphone of one earphone can reduce power consumption of the headset.
(203) In some embodiments, an image captured by a camera of the first electronic device **100** includes a plurality of faces. That is, a plurality of users jointly use the first electronic device **100** to perform video recording. One of the plurality of users wears the TWS headset. The TWS headset establishes a communication connection to the first electronic device **100**. The near field area of the first microphone array formed by the microphone of the first electronic device **100** and the microphone of the TWS headset may generally cover an area in which the plurality of users are located. In this case, a sound in the near field area of the first microphone array may include both voices of the plurality of users and sounds of playing musical instruments by the plurality of users. That is, when the first electronic device **100** performs spatial filtering on the joint microphone array signal collected by the first microphone array, the voice of the user wearing the TWS headset and the sound of playing a musical instrument by the user can be enhanced, and a voice of another user performing video recording and a sound of playing a musical instrument by the another user can be enhanced. This can improve sound quality of a video recorded by a plurality of persons.
(204) FIG. **8** shows an example of a schematic diagram of a structure of a first electronic device **100**.
(205) A first electronic device **100** may include a processor **110**, an external memory interface **120**, an internal memory **121**, a universal serial bus (universal serial bus, USB) interface **130**, a charging

management module **140**, a power management module **141**, a battery **142**, an antenna **1**, an antenna **2**, a mobile communication module **150**, a wireless communication module **160**, an audio module **170**, a speaker **170**A, a receiver **170**B, a microphone **170**C, a headset jack **170**D, a sensor module **180**, a button **190**, a motor **191**, an indicator **192**, a camera **193**, a display **194**, a subscriber identification module (subscriber identification module, SIM) card interface **195**, and the like. The sensor module **180** may include a pressure sensor **180**A, a gyroscope sensor **180**B, a barometric pressure sensor **180**C, a magnetic sensor **180**D, an acceleration sensor **180**E, a distance sensor **180**F, an optical proximity sensor **180**G, a fingerprint sensor **180**H, a temperature sensor **180**J, a touch sensor **180**K, an ambient light sensor **180**L, a bone conduction sensor **180**M, and the like.
(206) It can be understood that the structure shown in this embodiment of the present invention does not constitute a specific limitation on the first electronic device **100**. In some other embodiments of this application, the first electronic device **100** may include more or fewer components than those shown in the figure, or combine some components, or split some components, or have different component arrangements. The components shown in the figure may be implemented by hardware, software, or a combination of software and hardware.
(207) The processor **110** may include one or more processing units. For example, the processor **110** may include an application processor (application processor, AP), a modem processor, a graphics processing unit (graphics processing unit, GPU), an image signal processor (image signal processor, ISP), a controller, a memory, a video codec, a digital signal processor (digital signal processor, DSP), a baseband processor, and/or a neural-network processing unit (neural-network processing unit, NPU). Different processing units may be independent components, or may be integrated into one or more processors.
(208) A memory may be further disposed in the processor **110**, and is configured to store instructions and data. In some embodiments, the memory in the processor **110** is a cache memory. The memory may store an instruction or data that has been used or cyclically used by the processor **110**. If the processor **110** needs to use the instructions or the data again, the processor may directly invoke the instructions or the data. from the memory. This avoids repeated access, reduces waiting time of the processor **110**, and improves system efficiency.
(209) The charging management module **140** is configured to receive a charging input from a charger.
(210) The power management module **141** is configured to connect the battery **142** and the charging management module **140** to the processor **110**. The power management module **141** receives input of the battery **142** and/or the charging management module **140**, to supply power to the processor **110**, the internal memory **121**, an external memory, the display **194**, the camera **193**, the wireless communication module **160**, and the like.
(211) A wireless communication function of the first electronic device **100** may be implemented by using the antenna **1**, the antenna **2**, the mobile communication module **150**, the wireless communication module **160**, the modem processor, the baseband processor, and the like.
(212) The antenna **1** and the antenna **2** are configured to transmit and receive an electromagnetic wave signal. Each antenna in the first electronic device **100** may be configured to cover one or more communication bands.
(213) The mobile communication module **150** can provide a wireless communication solution that is applied to the first electronic device **100** and that includes 2G/3G/4G/5G or the like. The mobile communication module **150** may receive an electromagnetic wave through the antenna **1**, perform processing such as filtering or amplification on the received electromagnetic wave, and transmit the electromagnetic wave to the modem processor for demodulation. The mobile communication module **150** may further amplify a signal modulated by the modem processor, and convert the signal into an electromagnetic wave for radiation through the antenna **1**.
(214) The modem processor may include a modulator and a demodulator. The modulator is configured to modulate a to-be-sent low-frequency baseband signal into a medium-high frequency

signal. The demodulator is configured to demodulate a received electromagnetic wave signal into a low-frequency baseband signal. In some embodiments, the modem processor may be an independent component. In some other embodiments, the modem processor may be independent of the processor **110**, and is disposed in a same device as the mobile communication module **150** or another functional module.
(215) The wireless communication module **160** may provide a wireless communication solution that includes a wireless local area network (wireless local area networks, WLAN) (for example, a wireless fidelity (wireless fidelity, Wi-Fi) network), Bluetooth (Bluetooth, BT), a global navigation satellite system (global navigation satellite system, GNSS), frequency modulation (frequency modulation, FM), a near field communication (near field communication, NFC) technology, an infrared (infrared, IR) technology, or the like and that is applied to the first electronic device **100**. The wireless communication module **160** may be one or more components integrating at least one communications processor module. The wireless communication module **160** receives an electromagnetic wave through the antenna **2**, performs frequency modulation and filtering processing on an electromagnetic wave signal, and transmits a processed signal to the processor **110**. The wireless communication module **160** may further receive a to-be-sent signal from the processor **110**, perform frequency modulation and amplification on the signal, and convert a processed signal into an electromagnetic wave through the antenna **2** for radiation.
(216) The first electronic device **100** implements a display function by using the GPU. the display **194**, the application processor, and the like. The GPU is a microprocessor for image processing, and is connected to the display **194** and the application processor. The GPU is configured to: perform mathematical and geometric computation, and render an image. The processor **110** may include one or more GPUs, which execute program instructions to generate or change display information.
(217) The display **194** is configured to display an image, a video, and the like. In some embodiments, the first electronic device **100** may include one or N displays **194**, where N is a positive integer greater than 1.
(218) The first electronic device **100** may implement a photographing function by using the ISP, the camera **193**, the video codec, the GPU, the display **194**, the application processor, and the like.
(219) The ISP is configured to process data fed back the camera **193**. For example, during photographing, a shutter is pressed and light is transmitted to a photosensitive element of the camera through a lens. An optical signal is converted into an electrical signal, and the photosensitive element of the camera transmits the electrical signal to the ISP for processing, to convert the electrical signal into a visible image. In some embodiments, the ISP may be disposed in the camera **193**.
(220) The camera **193** is configured to capture a static image or a video. An optical image of an object is generated through the lens, and is projected onto the photosensitive element. The photosensitive element may be a charge coupled device (charge coupled device, CCD) or a complementary metal-oxide-semiconductor (complementary metal-oxide-semiconductor, CMOS) phototransistor. The light-sensitive element converts an optical signal into an electrical signal, and then transmits the electrical signal to the ISP to convert the electrical signal into a digital image signal. The ISP outputs the digital image signal to the DSP for processing. The DSP converts the digital image signal into an image signal in a standard format such as RGB or YUV. In some embodiments, the first electronic device **100** may include one or N cameras **193**, where N is a positive integer greater than 1.
(221) In this application, the camera **213** shown in FIG. **4** is the front camera in the camera **193** in FIG. **8**. An image collected by the camera **213** may be converted into a digital image signal by the ADC **215**C and output to the DSP. The ADC **215**C may be an analog to digital converter integrated into the ISP.
(222) The DSP is configured to process a digital signal, for example, a digital signal such as a

digital image signal or a digital audio signal. For example, when the first electronic device **100** selects a frequency point, the digital signal processor is configured to perform Fourier transformation on frequency energy.
(223) In some embodiments of this application, the DSP may include a signal generator, a delay alignment module, a coordinate resolving module, and a spatial filtering module. For functions of the signal generator, the delay alignment module, the coordinate resolving module, and the spatial filtering module, refer to the description of the embodiment in FIG. **4**. Details are not described herein again.
(224) The signal generator, the delay alignment module, the coordinate resolving module, and the spatial filtering module may be integrated into another chip processor separately or together, and are not limited to being integrated into the DSP. This is not limited in this embodiment of this application.
(225) The video codec is configured to compress or decompress a digital video. The first electronic device **100** may support one or more video codecs. In this way, the first electronic device **100** may play or record videos in a plurality of encoding formats, for example, moving picture experts group (moving picture experts group, MPEG)-1, MPEG-2, MPEG-3, and MPEG-4.
(226) The NPU is a neural-network (neural-network, NN) computing processor, quickly processes input information by referring to a structure of a biological neural network, for example, by referring to a mode of transmission between human brain neurons, and may further continuously perform self-learning. Applications such as intelligent cognition of the first electronic device **100** may be implemented through the NPU, for example, image recognition, facial recognition, speech recognition, and voice activity detection.
(227) The external memory interface **120** may be configured to connect to an external memory card, for example, a micro SD card, to expand a storage capability of the first electronic device **100**.
(228) The internal memory **121** may be configured to store computer-executable program code. The executable program code includes instructions. The processor **110** executes various function applications and data. processing of the first electronic device **100** by running the instructions stored in the internal memory **121**. The internal memory **121** may include a program storage area and a data storage area. The program storage area may store an operating system, an application required by at least one function (for example, a sound playing function or an image playing function), and the like. The data storage area may store data (such as audio data) created during use of the first electronic device **100**. In addition, the internal memory **121** may include a high-speed random access memory, and may further include a nonvolatile memory, for example, at least one magnetic disk storage device, a flash storage device, a universal flash storage (universal flash storage, UFS), and the like.
(229) The first electronic device **100** may implement audio functions such as music playing and recording by using the audio module **170**, the speaker **170**A, the receiver **170**B, the microphone **170**C, the headset jack **170**D, the application processor, and the like.
(230) The audio module **170** is configured to convert a digital audio signal into an analog audio signal for output, and is also configured to convert analog audio input into a digital audio signal. The audio module **170** may be further configured to encode and decode an audio signal. In some embodiments, the audio module **170** may be disposed in the processor **110**, or some function modules of the audio module **170** are disposed in the processor **110**.
(231) The speaker **170**A, also referred to as a “loudspeaker”, is configured to convert an audio electrical signal into a sound signal. The first electronic device **100** may be used to listen to music or answer a call in a hands-free mode through the speaker **170**A.
(232) The receiver **170**B, also referred to as an “earpiece”, is configured to convert an electrical audio signal into a sound signal. When a call is answered or voice information is received through the first electronic device **100**, the receiver **170**B may be put close to a human ear to listen to a

voice.
(233) The microphone **170**C, also referred to as a “mike” or a “mic”, is configured to convert a sound signal into an electrical signal. When making a call or sending a voice message, a user may make a sound near the microphone **1700** through the mouth of the user, to input a sound signal to the microphone **170**C. At least one microphone **170**C may be disposed in the first electronic device **100**. In some other embodiments, two microphones **170**C may be disposed in the first electronic device **100**, to collect a sound signal and implement a noise reduction function. In some other embodiments, three, four, or more microphones **170**C may alternatively be disposed in the first electronic device **100**, to collect a sound signal, implement noise reduction, and identify a sound source, to implement a directional recording function and the like.
(234) The microphone **170**C is the first microphone **211** in the foregoing embodiment.
(235) The headset jack **170**D is configured to connect to a wired headset.
(236) In this application, the signal generator of the first electronic device **100** may generate an alignment signal. The alignment signal is a digital audio signal. The signal generator may send the alignment signal to the DAC **215**A. The DAC **215**A may convert the alignment signal into an analog audio signal. The DAC **215**A may be integrated into the audio module **170**.
(237) A sound collected by the first microphone **211** is an analog audio signal. The first microphone **211** may send the collected sound to the ADC **215**B. The ADC **215**B may convert the analog audio signal into a digital audio signal. The ADC **215**B may be integrated into the audio module **170**.
(238) The pressure sensor **180**A is configured to sense a pressure signal, and can convert the pressure signal into an electrical signal. In some embodiments, the pressure sensor **180**A may be disposed on the display **194**.
(239) The gyroscope sensor **180**B may be configured to determine a moving posture of the first electronic device **100**. In some embodiments, an angular velocity of the first electronic device **100** around three axes (namely, axes x, y, and z) may be determined by using the gyroscope sensor **180**B. The gyroscope sensor **180**B may be configured to implement image stabilization during photographing. For example, when the shutter is pressed, the gyroscope sensor **180**B detects an angle at which the first electronic device **100** jitters, obtains, through calculation based on the angle, a distance for which a lens module needs to compensate, and allows the lens to cancel a jitter of the first electronic device **100** through reverse motion, to implement image stabilization. The gyroscope sensor **180**B may also be used in a navigation scenario and a somatic game scenario.
(240) The barometric pressure sensor **180**C is configured to measure barometric pressure.
(241) The magnetic sensor **180**D includes a Hall sensor. The first electronic device **100** may detect opening and closing of a flip cover by using the magnetic sensor **180**D.
(242) The acceleration sensor **180**E may detect accelerations in various directions (usually on three axes) of the first electronic device **100**. When the first electronic device **100** is still, a magnitude and a direction of gravity may be detected. The acceleration sensor may be further configured to identify a posture of the first electronic device **100**, and is applied to an application such as landscape/portrait mode switching and a pedometer.
(243) The gyroscope sensor **180**B and the acceleration sensor **180**E may be the posture sensor **214** in the foregoing embodiment.
(244) The distance sensor **180**F is configured to measure a distance. The first electronic device **100** may measure the distance through infrared or a laser. In some embodiments, in an image shooting scenario, the first electronic device **100** may measure a distance by using the distance sensor **180**F to implement quick focusing.
(245) The optical proximity sensor **180**G may include, for example, a light-emitting diode (LED), and an optical detector, for example, a photodiode. The light emitting diode may be an infrared light emitting diode. The first electronic device **100** emits infrared light through the light-emitting diode. The first electronic device **100** detects infrared reflected light from a nearby object through the photodiode. When detecting sufficient reflected light, the first electronic device **100** may

determine that there is an object near the first electronic device **100**. When detecting insufficient reflected light, the first electronic device **100** may determine that there is no object near the first electronic device **100**. The first electronic device **100** may detect, through the optical proximity sensor **180**G, that the user holds the first electronic device **100** close to an ear during a call, so that the first electronic device **100** automatically turns off a screen for power saving.
(246) The ambient light sensor **180**L is configured to sense ambient light brightness. The first electronic device **100** may adaptively adjust brightness of the display **194** based on the sensed ambient light brightness.
(247) The fingerprint sensor **180**H is configured to collect a fingerprint. The first electronic device **100** may use a feature of the collected fingerprint to implement fingerprint-based unlocking, application lock accessing, fingerprint-based photographing, fingerprint-based call answering, and the like.
(248) The temperature sensor **180**J is configured to detect a temperature. In some embodiments, the first electronic device **100** executes a temperature processing policy based on the temperature detected by the temperature sensor **180**J.
(249) The touch sensor **180**K is also referred to as a “touch panel”. The touch sensor **180**K may be disposed on the display **194**, and the touch sensor **180**K and the display **194** constitute a touchscreen, which is also referred to as a “touchscreen”. The touch sensor **180**K is configured to detect a touch operation performed on or near the touch sensor **180**K. The touch sensor may transfer the detected touch operation to the application processor to determine a type of the touch event. A visual output related to the touch operation may be provided on the display **194**.
(250) The bone conduction sensor **180**M may obtain a vibration signal.
(251) The button **190** includes a power button, a volume button, and the like. The button **190** may be a mechanical button, or may be a touch button.
(252) The motor **191** may generate a vibration prompt.
(253) The indicator **192** may be an indicator light, and may be configured to indicate a charging status and a power change, or may be configured to indicate a message, a missed call, a notification, and the like.
(254) The SIM card interface **195** is configured to connect to a SIM card. The SIM card may be inserted into the SIM card interface **195** or removed from the SIM card interface **195**, to implement contact with and separation from the first electronic device **100**.
(255) In conclusion, the foregoing embodiments are merely intended for describing the technical solutions of this application, but not for limiting this application. Although this application is described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should understand that they may still make modifications to the technical solutions described in the foregoing embodiments or make equivalent replacements to some technical features thereof, without departing from the scope of the technical solutions of embodiments of this application.

Claims

1. A method implemented by a first electronic device, wherein the method comprises: collecting a face image; determining relative locations of a first microphone, a second microphone, and a third microphone of the first electronic device based on the face image and posture information of the first electronic device; obtaining a first audio signal of the first microphone, a second audio signal of the second microphone, and a third audio signal of the third microphone; and performing noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations.
2. The method according to claim 1, wherein, before the performing the noise reduction processing, the method further comprises: generating an alignment signal; performing digital-to-analog conversion on the alignment signal to obtain an alignment sound; emitting the alignment sound;

further obtaining the first audio signal, the second audio signal, and the third audio signal by detecting a sound, wherein the sound includes the alignment sound; performing delay correlation detection on a first alignment signal part of the first audio signal, a second alignment signal part of the second audio signal, and a third alignment signal part of the third audio signal to determine delay lengths between the first audio signal, the second audio signal, and the third audio signal; and performing delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the delay lengths.
3. The method according to claim 2, wherein the alignment signal is an audio signal with a frequency higher than 20000 Hertz (Hz).
4. The method according to claim 1, wherein determining the relative locations comprises: determining a standard head coordinate system based on a standard head model; storing coordinates in key points of the standard head model in the standard head coordinate system; determining a first conversion relationship between the standard head coordinate system and a first electronic device coordinate system based on a correspondence between first coordinates of a first human face key point in the standard head coordinate system and second coordinates of the first human face key point in a face image coordinate system; determining third coordinates of a left ear and a right ear in the standard head model in the first electronic device coordinate system based on the first conversion relationship and fourth coordinates of the left ear and the right ear in the standard head model in the standard head coordinate system, wherein the third coordinates are respectively fifth coordinates of the second microphone and the third microphone in the first electronic device coordinate system; determining a second conversion relationship between the first electronic device coordinate system and a world coordinate system based on the posture information; and determining the relative locations based on the second conversion relationship and seventh coordinates of the first microphone, the second microphone, and the third microphone in the first electronic device coordinate system, wherein the relative locations comprise eighth coordinates of the first microphone, the second microphone, and the third microphone in the world coordinate system.
5. The method according to claim 1, wherein performing the noise reduction processing comprises: performing voice activity detection on the first audio signal, the second audio signal, and the third audio signal based on the relative locations; determining, using the voice activity detection, a first frequency point of a target sound signal and a second frequency point of an ambient noise signal in the first audio signal, the second audio signal, and the third audio signal, wherein the target sound signal is a sound signal of a sound source located in a near field area of a microphone array formed by the first microphone, the second microphone, and the third microphone; updating a noise spatial characteristic of ambient noise based on the first frequency point and the second frequency point, wherein the noise spatial characteristic indicates a spatial distribution of the ambient noise, and wherein the spatial distribution comprises a direction and energy of the ambient noise; determining a target steering vector of the first audio signal, the second audio signal, and the third audio signal based on the relative locations, wherein the target steering vector indicates a direction of the target sound signal; determining a spatial filter based on the noise spatial characteristic and the target steering vector; and performing spatial filtering on the first audio signal, the second audio signal, and the third audio signal using the spatial filter.
6. The method according to claim 1, wherein the method further comprises: performing, using a left-ear wireless earphone of the first electronic device, first wearing detection to determine whether the left-ear wireless earphone is in an in-ear state; obtaining, when the left-ear wireless earphone is in the in-ear state, the second audio signal using the second microphone; performing, using a right-ear wireless earphone, second wearing detection to determine whether the right-ear wireless earphone is in an in-ear state; and obtaining, when the right-ear wireless earphone is in the in-ear state, the third audio signal using the third microphone.
7. The method according to claim 1, wherein the method further comprises: obtaining the first

audio signal, the second audio signal, and the third audio signal in a first time period using the first microphone, the second microphone, and the third microphone, respectively; obtaining a fourth audio signal after performing the noise reduction processing; and mixing a first video and the fourth audio signal that are collected in the first time period.

8. A sound collecting method performed by a first electronic device wherein the method comprises: collecting face image; determining relative locations of a first microphone, a second microphone, and a third microphone of the first electronic device based on the face image and posture information of the first electronic device; obtaining a first audio signal of the first microphone, a second audio signal of the second microphone, and a third audio signal of the third microphone; generating an alignment signal; performing a delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the alignment signal; and further performing noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations.

9. The sound collecting method according to claim 8, wherein performing the delay alignment comprises: performing digital-to-analog conversion on the alignment signal to obtain an alignment sound; emitting the alignment sound; further obtaining the first audio signal, the second audio signal, and the third audio signal by detecting a sound, wherein the sound includes the alignment sound; performing a delay correlation detection on a first alignment signal part of the first audio signal, a second alignment signal part of the second audio signal, and a third alignment signal part of the third audio signal, to determine delay lengths between the first audio signal, the second audio signal, and the third audio signal; and performing a delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the delay lengths.

10. The sound collecting method according to claim 9, wherein the alignment signal is an audio signal with a frequency higher than 20000 Hertz (Hz).

11. The sound collecting method according to claim 8, wherein determining the relative locations comprises: determining a standard head coordinate system based on a standard head model; storing coordinates in key points of the standard head model in the standard head coordinate system; determining a first conversion relationship between the standard head coordinate system and a first electronic device coordinate system based on a correspondence between first coordinates of a first human face key point in the standard head coordinate system and second coordinates of the first human face key point in a face image coordinate system; determining third coordinates of a left ear and a right ear in the standard head model in the first electronic device coordinate system based on the first conversion relationship and fourth coordinates of the left ear and the right ear in the standard head model in the standard head coordinate system, wherein the third coordinates are respectively fifth coordinates of the second microphone and the third microphone in the first electronic device coordinate system; determining a second conversion relationship between the first electronic device coordinate system and a world coordinate system based on the posture information; and determining the relative locations based on the second conversion relationship and seventh coordinates of the first microphone, the second microphone, and the third microphone in the first electronic device coordinate system, wherein the relative locations comprise eighth coordinates of the first microphone, the second microphone, and the third microphone in the world coordinate system.

12. The sound collecting method according to claim 8, wherein performing the noise reduction processing comprises: performing voice activity detection on the first audio signal, the second audio signal, and the third audio signal based on the relative locations; determining, using the voice activity detection, a first frequency point of a target sound signal and a second frequency point of an ambient noise signal in the first audio signal, the second audio signal, and the third audio signal, wherein the target sound signal is a sound signal of a sound source located in a near field area of a microphone array formed by the first microphone, the second microphone, and the third microphone; updating a noise spatial characteristic of ambient noise based on the first frequency

point and the second frequency point, wherein the noise spatial characteristic indicates a spatial distribution of the ambient noise, and wherein the spatial distribution comprises a direction and energy of the ambient noise; determining a target steering vector of the first audio signal, the second audio signal, and the third audio signal based on the relative locations, wherein the target steering vector indicates a direction of the target sound signal; determining a spatial filter based on the noise spatial characteristic and the target steering vector; and performing spatial filtering on the first audio signal, the second audio signal, and the third audio signal using the spatial filter.
13. The sound collecting method according to claim 8, wherein the method further comprises: obtaining the first audio signal, the second audio signal, and the third audio signal in a first time period using the first microphone, the second microphone, and the third microphone, respectively; obtaining a fourth audio signal after performing the noise reduction processing; and mixing a first video and the fourth audio signal that are collected in the first time period.
14. The sound collecting method according to claim 8, wherein the delay alignment is performed before the noise reduction processing on the first audio signal, the second audio signal, and the third audio signal.
15. The sound collecting method according to claim 8, wherein the method further comprises: performing, using a left-ear wireless earphone of the first electronic device, wearing detection to determine whether the left-ear wireless earphone is in an in-ear state; obtaining, when the left-ear wireless earphone is in the in-ear state, the second audio signal using the second microphone; performing, using a right-ear wireless earphone, second wearing detection to determine whether the right-ear wireless earphone is in an in-ear state; and obtaining, when the right-ear wireless earphone is in the in-ear state, the third audio signal using the third microphone.
16. An electronic device comprising: a memory configured to store computer instructions; and a processor coupled to the memory and configured to invoke the computer instructions to cause the electronic device to: collect a face image; determine relative locations of a first microphone, a second microphone, and a third microphone based on the face image and posture information of the first electronic device; obtain a first audio signal of the first microphone, a second audio signal of the second microphone, and a third audio signal of the third microphone; and perform noise reduction processing on the first audio signal, the second audio signal, and the third audio signal based on the relative locations.
17. The electronic device according to claim 16, wherein, when executed by the processor, the computer instructions further cause the electronic device to: generate an alignment signal; perform digital-to-analog conversion on the alignment signal to obtain an alignment sound; emit the alignment sound; and further obtain the first audio signal, the second audio signal, and the third audio signal by detecting a sound, wherein the sound includes the alignment sound; perform delay correlation detection on a first alignment signal part of the first audio signal, a second alignment signal part of the second audio signal, and a third alignment signal part of the third audio signal to determine delay lengths between the first audio signal, the second audio signal, and the third audio signal; and perform delay alignment on the first audio signal, the second audio signal, and the third audio signal based on the delay lengths.
18. The electronic device according to claim 16, wherein, when executed by the processor, the computer instructions further cause the electronic device to: obtain the first audio signal, the second audio signal, and the third audio signal in a first time period using the first microphone, the second microphone, and the third microphone, respectively; obtain a fourth audio signal after performing the noise reduction processing; and mix a first video and the fourth audio signal that are collected in the first time period.
19. The electronic device according to claim 16, wherein the computer instructions to perform the noise reduction processing further comprises instructions, when executed by the processor, cause the electronic device to: perform voice activity detection on the first audio signal, the second audio signal, and the third audio signal based on the relative locations; determine, using the voice activity

detection, a first frequency point of a target sound signal and a second frequency point of an ambient noise signal in the first audio signal, the second audio signal, and the third audio signal, wherein the target sound signal is a sound signal of a sound source located in a near field area of a microphone array formed by the first microphone, the second microphone, and the third microphone; and update a noise spatial characteristic of ambient noise based on the first frequency point and the second frequency point, wherein the noise spatial characteristic indicates a spatial distribution of the ambient noise, and wherein the spatial distribution comprises a direction and energy of the ambient noise.

20. The electronic device according to claim 19, wherein the computer instructions to perform the noise reduction processing further comprises instructions, when executed by the processor, cause the electronic device to: determine a target steering vector of the first audio signal, the second audio signal, and the third audio signal based on the relative locations, wherein the target steering vector indicates a direction of the target sound signal; determine a spatial filter based on the noise spatial characteristic and the target steering vector; and perform spatial filtering on the first audio signal, the second audio signal, and the third audio signal using the spatial filter.